Stochastic Calculus and Barrier Options: A Black-Scholes Extension

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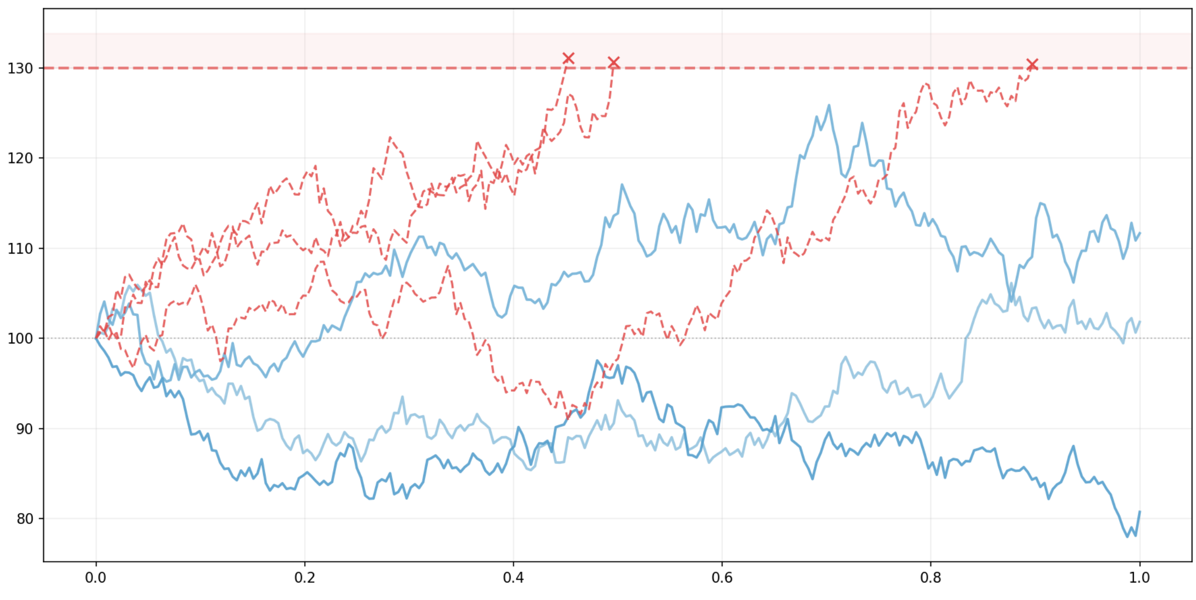

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Abstract

This report presents the Black–Scholes framework and its extension to path-dependent derivatives. Starting from the properties of Brownian motion, we introduce the essential tools of stochastic calculus, including Girsanov's theorem for the change of measure and the resulting risk-neutral valuation. We derive the Black–Scholes PDE and its closed-form solution for European options, demonstrating the equivalence between replication-based and probabilistic approaches. A significant focus is placed on the Reflection Principle, which allows for the joint distribution of the Brownian motion and its running maximum.These techniques are then applied to the pricing of different types of barrier options, where we recalculate the conditional expectations and establish specific boundary conditions for the underlying PDE. Furthermore, we implement numerical simulations using real market data to evaluate the behavior of these instruments. Finally, we look at the model's limitations, focusing on the constant volatility and interest rate assumptions, before briefly exploring more complex models.

Contents

Introduction

This report is submitted as part of the Projet Recherche et Innovation (PRI) within the Department of Mathematical and Modeling Engineering (GMM) at INSA Toulouse. It serves as a direct continuation of the work initiated during the first semester's *Reading Seminar*, which focused on the foundational Black–Scholes model.
Building upon our previous study of European options, this project aims to extend the Black–Scholes framework [1] to more complex, path-dependent derivatives. While the standard model relies on terminal asset values, the valuation of instruments such as barrier options requires [2] a deeper exploration of the trajectory of the underlying asset.
To achieve this, we first establish the essential tools of stochastic calculus, notably Itô's lemma and Girsanov's theorem for the change of measure and the resulting risk-neutral valuation. We then bridge the gap between replication-based and probabilistic approaches by deriving the Black–Scholes PDE and its closed-form solutions. A significant portion of this work is dedicated to the Reflection Principle, which provides the mathematical key to determining the joint distribution of Brownian motion and its running maximum.
Finally, these techniques are applied to price various barrier options. This theoretical framework is complemented by numerical simulations using real market data to evaluate the practical behavior of these derivatives. The report concludes with a discussion on the limitations of the Black–Scholes model, particularly regarding the assumption of constant volatility and interest rate, before exploring more complex models [3] [4] .

1 Stochastic Calculus and Measure Transformation

This section details the fundamental properties of Brownian motion and the Girsanov theorem, which allows for the change of probability measures in stochastic modeling

1.1 Properties and Martingales of Brownian Motion

Standard Brownian Motion

Definition 1. A real-valued stochastic process $(B_t)_{t\geq 0}$ is a standard Brownian motion if:

1. $B_0 = 0$ a.s. and the paths $t \mapsto B_t$ are almost surely continuous;
2. $(B_t)_{t\geq 0}$ is a centered Gaussian process with $\mathrm{Cov}(B_s, B_t) = \min(s,t)$.

This covariance structure implies $\mathrm{Var}(B_t) = t$. While (B_t) is the continuous limit of a random walk, it is uniquely characterized by having independent and stationary increments: for $0 \leq s < t$, $B_t - B_s \sim \mathcal{N}(0, t-s)$ and is independent of $\mathcal{F}_s \coloneqq \sigma(B_u : u \leq s)$.

Joint Density. For $0 < s < t$, (B_s, B_t) is a Gaussian vector with mean $\mathbf{0}$ and covariance $\Sigma = \left(\begin{smallmatrix} s & s \\ s & t \end{smallmatrix}\right)$. With $\det(\Sigma) = s(t-s)$, its joint density is:

$$f_{(B_s,B_t)}(x,y) = \frac{1}{2\pi\sqrt{s(t-s)}} \exp\left(-\frac{1}{2}\left[\frac{x^2}{s} + \frac{(y-x)^2}{t-s}\right]\right). \tag{1}$$

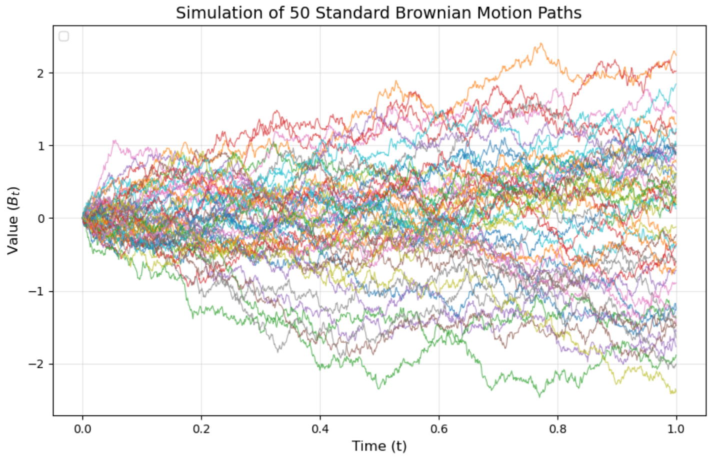

Figure 1: Numerical simulation of Brownian motion paths.

Fundamental Brownian Martingales

Martingales are the primary tools for risk-neutral valuation. Let $\mathcal{F}_t$ be the natural filtration of B. We identify three fundamental $(\mathcal{F}_t, \mathbb{P})$-martingales:

1. **Brownian Motion:** $(B_t)_{t\geq 0}$ is a martingale. Using $B_t = B_s + (B_t - B_s)$, the independence and centered nature of increments yield $\mathbb{E}[B_t \mid \mathcal{F}_s] = B_s + \mathbb{E}[B_t - B_s] = B_s$.

2. **Quadratic Martingale:** $(B_t^2 - t)_{t\geq 0}$ is a martingale. Expanding $B_t^2 = (B_s + (B_t - B_s))^2$ and taking the conditional expectation:

$$\mathbb{E}[B_t^2 \mid \mathcal{F}_s] = B_s^2 + 2B_s\mathbb{E}[B_t - B_s] + \mathbb{E}[(B_t - B_s)^2] = B_s^2 + (t-s).$$

   Subtracting t on both sides recovers the martingale property.

3. **Exponential Martingale:** For $\theta \in \mathbb{R}$, $M_t^\theta := \exp(\theta B_t - \frac{1}{2}\theta^2 t)$ is a martingale. Defining $X = B_t - B_s \sim \mathcal{N}(0, t-s)$, we use the Gaussian moment generating function $\mathbb{E}[e^{\theta X}] = e^{\frac{1}{2}\theta^2(t-s)}$:

$$\mathbb{E}[M_t^\theta \mid \mathcal{F}_s] = M_s^\theta \cdot e^{-\frac{1}{2}\theta^2(t-s)}\mathbb{E}[e^{\theta(B_t - B_s)} \mid \mathcal{F}_s] = M_s^\theta \cdot e^{-\frac{1}{2}\theta^2(t-s)}e^{\frac{1}{2}\theta^2(t-s)} = M_s^\theta.$$

$\square$

1.2 Itô's Lemma and Stochastic Calculus

The fundamental difference between classical calculus and stochastic calculus lies in the non-zero quadratic variation of Brownian motion ($d\langle B, B\rangle_t = dt$). This necessitates a second-order expansion to differentiate functions of stochastic processes.

Itô's Formula for Brownian Motion Let $f \in \mathcal{C}^{2,1}(\mathbb{R} \times \mathbb{R}_+)$. The differential of the process $f(B_t, t)$ is given by:

$$df(B_t, t) = \left(\partial_t f + \frac{1}{2}\partial^2_{xx} f\right) dt + \partial_x f dB_t. \tag{2}$$

Generalization to an SDE In financial modeling, we primarily work with processes $(X_t)_{t\geq 0}$ that are solutions to a Stochastic Differential Equation (SDE) of the form:

$$dX_t = \mu_t dt + \sigma_t dB_t. \tag{3}$$

Applying Itô's formula to $f(X_t, t)$, we obtain the general form used for pricing:

$$df(X_t, t) = \left(\partial_t f + \mu_t \partial_x f + \frac{1}{2}\sigma_t^2 \partial^2_{xx} f\right) dt + \sigma_t \partial_x f dB_t. \tag{4}$$

Remark 1. A process $(f(X_t, t))_{t\geq 0}$ is a local martingale if and only if its drift term vanishes. This condition leads directly to the Black–Scholes Partial Differential Equation (PDE) when applied to the discounted asset price.

1.3 Girsanov's Theorem and Change of Measure

To transition from a physical description of price dynamics to a risk-neutral pricing framework, we require a mechanism to modify the drift of a process while preserving its Brownian structure. This is achieved through a change of probability measure.

The Radon-Nikodym Process

Let $(\mathcal{F}_t)_{t\in[0,T]}$ be the natural filtration. We define a new probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ on $\mathcal{F}_T$ via the density $Z_T = \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}$. For any $\mathcal{F}_t$-measurable random variable L_t, the expectation under $\mathbb{Q}$ is related to $\mathbb{P}$ by:

$$\mathbb{E}_{\mathbb{Q}}[L_t] = \mathbb{E}_{\mathbb{P}}[L_t Z_T] = \mathbb{E}_{\mathbb{P}}\left[\mathbb{E}_{\mathbb{P}}[L_t Z_T \mid \mathcal{F}_t]\right] = \mathbb{E}_{\mathbb{P}}[L_t Z_t],$$

where $Z_t = \mathbb{E}_{\mathbb{P}}[Z_T \mid \mathcal{F}_t]$ is the density process. If we define Z as a stochastic exponential of the form $\mathrm{d}Z_t = Z_t u_t \, \mathrm{d}B_t$, then Z is a $\mathbb{P}$-martingale:

$$Z_t = \exp\left(\int_0^t u_s \, \mathrm{d}B_s - \frac{1}{2}\int_0^t u_s^2 \, \mathrm{d}s\right).$$

Girsanov's Theorem

The following theorem formalizes how the drift of a Brownian motion is transformed under the measure $\mathbb{Q}$ induced by the density process Z_t.

Theorem 1 (Girsanov)**.** *Let $(B_t)_{t\geq 0}$ be a standard Brownian motion under $\mathbb{P}$ and $(u_t)_{t\geq 0}$ an adapted process. Define the process $(\widetilde{B}_t)_{t\geq 0}$ by:*

$$\widetilde{B}_t \coloneqq B_t - \int_0^t u_s \, \mathrm{d}s. \tag{5}$$

Then, under the probability measure $\mathbb{Q}$ defined by the Radon-Nikodym derivative $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\big|_{\mathcal{F}_t} = Z_t$, the process $(\widetilde{B}_t)_{t\geq 0}$ is a standard Brownian motion.

This result is fundamental in finance: it allows us to "absorb" the drift of an asset price into the Brownian motion by shifting to a measure $\mathbb{Q}$ where the discounted asset becomes a martingale.

2 The Black-Scholes Framework

The primary objective of this section is to synthesize the probabilistic tools previously introduced—specifically Brownian motion, martingales, and Girsanov's theorem—to derive the Black-Scholes pricing framework. We bridge the gap between stochastic dynamics and the valuation of contingent claims through the risk-neutral measure and the resulting Partial Differential Equation (PDE).

2.1 Market Dynamics and Risk-Neutral Valuation

Asset Dynamics and Stochastic Perturbation

In the Black-Scholes model, the evolution of a risky asset price $(X_t)_{t\geq 0}$ is governed by a Geometric Brownian Motion (GBM), expressed as the following Stochastic Differential Equation (SDE):

$$\mathrm{d}X_t = \mu X_t \, \mathrm{d}t + \sigma X_t \, \mathrm{d}B_t, \tag{6}$$

where $\mu \in \mathbb{R}$ represents the drift and $\sigma > 0$ is the volatility. Formally, setting $\sigma = 0$ reduces the model to a deterministic ODE with the solution $X_t = X_0 e^{\mu t}$. The SDE

(6) thus interprets market movements as a deterministic trend perturbed by continuous random noise, capturing the inherent uncertainty of price fluctuations.

The Risk-Neutral Measure $\mathbb{Q}$

A central tenet of arbitrage pricing theory is the existence of an equivalent martingale measure $\mathbb{Q}$ under which the discounted asset price is a martingale. By applying Girsanov's theorem with the kernel $u_t = \frac{\mu - r}{\sigma}$, we transition from the physical measure $\mathbb{P}$ to the risk-neutral measure $\mathbb{Q}$. Under $\mathbb{Q}$, the asset dynamics become:

$$\mathrm{d}X_t = rX_t\,\mathrm{d}t + \sigma X_t\,\mathrm{d}\widetilde{B}_t, \tag{7}$$

where r is the risk-free rate and $\widetilde{B}_t$ is a $\mathbb{Q}$-Brownian motion. This transformation is crucial as it ensures that the fair price of any derivative is independent of the investor's risk preference (represented by μ).

Option Payoff and Pricing Functional

We consider a European call option with maturity $T > 0$ and strike price $K > 0$. Its payoff at maturity is $g(X_T) = (X_T - K)^+$. The fair value of this claim at time $t \in [0, T]$, given $X_t = x$, is defined by the discounted conditional expectation under $\mathbb{Q}$:

$$V(t, x) := e^{-r(T-t)}\,\mathbb{E}^{\mathbb{Q}}\left[\,(X_T - K)^+ \,\middle|\, X_t = x\right]. \tag{8}$$

The following subsections will demonstrate how this probabilistic representation leads to the Black-Scholes PDE and the explicit pricing formula.

2.2 Derivation of the Black–Scholes PDE by Replication

In this section, we derive the Black–Scholes partial differential equation (PDE) using a replication argument. The key idea is to construct a self-financing portfolio that replicates the payoff of a European claim. By absence of arbitrage, the replicating portfolio must earn the risk-free rate once the stochastic risk has been hedged away.

Dynamics and Itô's Expansion

Assume that under the physical measure $\mathbb{P}$ the risky asset satisfies

$$\mathrm{d}X_t = \mu X_t\ \mathrm{d}t + \sigma X_t\ \mathrm{d}B_t, \qquad X_0 = x > 0. \tag{9}$$

Let $T > 0$ and let $g : \mathbb{R}_+ \to \mathbb{R}$ be a payoff function $g(x) = (x - K)^+$. We denote by $V(t, x)$ the time-t value of the claim. We assume that $V \in C^{1,2}([0, T) \times \mathbb{R}_+)$, so that Itô's lemma applies to the process $V(t, X_t)$.

Throughout this section, subscripts denote partial derivatives: $V_t = \frac{\partial V}{\partial t}$, $V_x = \frac{\partial V}{\partial x}$, and $V_{xx} = \frac{\partial^2 V}{\partial x^2}$. In finance, V_x corresponds to the *Delta* hedge ratio and V_{xx} to the *Gamma*. By Itô's formula:

$$\mathrm{d}V(t, X_t) = V_t(t, X_t)\ \mathrm{d}t + V_x(t, X_t)\ \mathrm{d}X_t + \frac{1}{2} V_{xx}(t, X_t)\ \mathrm{d}\langle X, X\rangle_t.$$

Since $\mathrm{d}\langle X, X\rangle_t = \sigma^2 X_t^2\ \mathrm{d}t$, we have:

$$\mathrm{d}V(t, X_t) = \Big(V_t + \mu X_t V_x + \tfrac{1}{2}\sigma^2 X_t^2 V_{xx}\Big)\ \mathrm{d}t + \sigma X_t V_x\ \mathrm{d}B_t. \tag{10}$$

Building the Replicating Portfolio

Consider a self-financing portfolio holding Δ_t units of X_t. Let Π_t be the portfolio value:

$$\mathrm{d}\Pi_t = \Delta_t\ \mathrm{d}X_t + r(\Pi_t - \Delta_t X_t)\ \mathrm{d}t. \tag{11}$$

To replicate the claim, we set $\Pi_t = V(t, X_t)$ and choose $\Delta_t := V_x(t, X_t)$. Substituting (9) into (11):

$$\mathrm{d}\Pi_t = \Big(\mu X_t V_x + r(V - X_t V_x)\Big)\ \mathrm{d}t + \sigma X_t V_x\ \mathrm{d}B_t. \tag{12}$$

The Black–Scholes PDE

Comparing (12) with (10), the diffusion terms coincide. For the replication to hold, the drifts must also match:

$$\Big(V_t + \mu X_t V_x + \tfrac{1}{2}\sigma^2 X_t^2 V_{xx}\Big) = \Big(\mu X_t V_x + r(V - X_t V_x)\Big).$$

The $\mu X_t V_x$ terms cancel, yielding the **Black–Scholes PDE**:

$$\boxed{V_t(t, x) + \frac{1}{2}\sigma^2 x^2 V_{xx}(t, x) + r x V_x(t, x) - r V(t, x) = 0} \tag{13}$$

with $V(T, x) = g(x)$. Notably, the drift μ disappears, confirming that prices are determined under risk-neutral valuation.

2.3 Risk-Neutral Valuation and Explicit Formula

Solution of the geometric Brownian motion

Applying Itô's formula to $\log X_t$ for (9):

$$\mathrm{d}(\log X_t) = \frac{1}{X_t}\ \mathrm{d}X_t - \frac{1}{2X_t^2}\ \mathrm{d}\langle X, X\rangle_t = \left(\mu - \frac{\sigma^2}{2}\right)\ \mathrm{d}t + \sigma\ \mathrm{d}B_t.$$

Integrating from 0 to t yields:

$$X_t = x_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right) t + \sigma B_t\right). \tag{14}$$

Explicit Derivation of the Black–Scholes Formula

Under the risk-neutral measure $\mathbb{Q}$, the asset price follows the stochastic differential equation $dX_t = rX_t\,dt + \sigma X_t\,dB_t$. The solution at maturity T is given by:

$$X_T = x_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right) T + \sigma B_T\right). \tag{15}$$

The value of a European call option at time 0 is the discounted expectation of its payoff:

$$\begin{aligned} C_0 &= e^{-rT}\mathbb{E}^{\mathbb{Q}}[(X_T - K)^+] \\ &= e^{-rT}\mathbb{E}^{\mathbb{Q}}[(X_T - K)\mathbf{1}_{\{X_T \geq K\}}] \\ &= e^{-rT}\mathbb{E}^{\mathbb{Q}}[X_T\mathbf{1}_{\{X_T \geq K\}}] - e^{-rT}K\mathbb{Q}(X_T \geq K). \end{aligned}$$

Step 1: Evaluation of the Probability Term $\mathbb{Q}(X_T \geq K)$.
Let $Z = \frac{B_T}{\sqrt{T}} \sim \mathcal{N}(0,1)$. the condition $X_T \geq K$ is equivalent to:

$$\begin{aligned} x_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right) T + \sigma\sqrt{T}Z\right) &\geq K \\ \left(r - \frac{\sigma^2}{2}\right) T + \sigma\sqrt{T}Z &\geq \ln\left(\frac{K}{x_0}\right) \\ Z &\geq \frac{\ln(K/x_0) - (r - \sigma^2/2)T}{\sigma\sqrt{T}} =: -d_2. \end{aligned}$$

where $d_2 = \frac{\ln(x_0/K) + (r-\sigma^2/2)T}{\sigma\sqrt{T}}$. Since the normal distribution is symmetric, $\mathbb{Q}(Z \geq -d_2) = \mathbb{Q}(Z \leq d_2) = \Phi(d_2)$. Thus, the second term is $-Ke^{-rT}\Phi(d_2)$.
Step 2: Evaluation of the Asset Term $\mathbb{E}^{\mathbb{Q}}[X_T\mathbf{1}_{\{X_T \geq K\}}]$.
Using the density of the standard normal distribution $\phi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$:

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[X_T\mathbf{1}_{\{X_T \geq K\}}] &= \int_{-d_2}^{+\infty} x_0 e^{(r-\sigma^2/2)T + \sigma\sqrt{T}x} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= \frac{x_0 e^{(r-\sigma^2/2)T}}{\sqrt{2\pi}} \int_{-d_2}^{+\infty} \exp\left(\sigma\sqrt{T}x - \frac{x^2}{2}\right) dx. \end{aligned}$$

To solve the integral, we complete the square in the exponent:

$$-\frac{1}{2}x^2 + \sigma\sqrt{T}x = -\frac{1}{2}(x^2 - 2\sigma\sqrt{T}x) = -\frac{1}{2}(x - \sigma\sqrt{T})^2 + \frac{1}{2}\sigma^2 T.$$

Substituting this back into the integral:

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[X_T \mathbf{1}_{\{X_T \geq K\}}] &= \frac{x_0 e^{(r-\sigma^2/2)T} e^{\frac{1}{2}\sigma^2 T}}{\sqrt{2\pi}} \int_{-d_2}^{+\infty} e^{-\frac{1}{2}(x-\sigma\sqrt{T})^2} dx \\ &= x_0 e^{rT} \int_{-d_2}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-\sigma\sqrt{T})^2} dx.\end{aligned}$$

We perform the change of variable $y = x - \sigma\sqrt{T}$, which gives $dy = dx$. The new lower bound is $-d_2 - \sigma\sqrt{T}$:

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[X_T \mathbf{1}_{\{X_T \geq K\}}] &= x_0 e^{rT} \int_{-d_2-\sigma\sqrt{T}}^{+\infty} \phi(y) dy \\ &= x_0 e^{rT} \Phi(d_2 + \sigma\sqrt{T}) \\ &= x_0 e^{rT} \Phi(d_1).\end{aligned}$$

where $d_1 = d_2 + \sigma\sqrt{T} = \frac{\ln(x_0/K)+(r+\sigma^2/2)T}{\sigma\sqrt{T}}$.

Final Result:

Discounting the two terms back to time 0, we obtain the Black–Scholes formula:

$$\boxed{C_0 = x_0 \Phi(d_1) - K e^{-rT} \Phi(d_2)} \tag{16}$$

Probabilistic Representation and Time-Homogeneity

To compute $V(t, x)$, we rely on the risk-neutral representation. Let $\tau = T - t$ be the time to maturity. We aim to show that $V(t, x)$ depends only on τ and x:

$$V(t, x) = e^{-r\tau} \mathbb{E}^{\mathbb{Q}} \left[(x\mathcal{X}_\tau - K)^+ \right].$$

Under $\mathbb{Q}$, $X_T = X_t \exp\left((r - \frac{1}{2}\sigma^2)\tau + \sigma(\tilde{B}_T - \tilde{B}_t) \right)$. By the stationarity and independence of Brownian increments, $(\tilde{B}_T - \tilde{B}_t)$ is independent of $\mathcal{F}_t$ and distributed as $\tilde{B}_\tau$. Defining $\mathcal{X}_\tau := \exp((r - \frac{1}{2}\sigma^2)\tau + \sigma\tilde{B}_\tau)$, the Markov property of X_t ensures:

$$\mathbb{E}^{\mathbb{Q}} \left[(X_T - K)^+ \mid \mathcal{F}_t \right] = \mathbb{E}^{\mathbb{Q}} \left[(X_t \mathcal{X}_\tau - K)^+ \mid X_t \right].$$

Conditioning on $X_t = x$, we confirm $V(t, x) = e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[(x\mathcal{X}_\tau - K)^+]$. This time-homogeneity proves that the no-arbitrage price coincides with the unique solution of the Black–Scholes PDE (13), bridging the replication and probabilistic approaches.

3 Reflection Principle and Stopping Time

While European options only depend on the terminal value S_T, barrier options are path-dependent. To price them, we must transition from marginal distributions to trajectory-based analysis using stopping times and the Reflection Principle. This section provides the stochastic foundations necessary to analyze the path-dependent nature of barrier options.

3.1 Stopping Time

The primary objective of introducing stopping times is to extend stochastic analysis by replacing deterministic time horizons with random variables τ

Definition 2 (Stopping Time). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space equipped with a filtration $(\mathcal{F}_t)_t \geq 0$. A random variable $\tau : \Omega \to [0, \infty]$ is called a stopping time with respect to the filtration $(\mathcal{F}_t)$ if, for every $t \geq 0$:

$$\{\tau \leq t\} \in \mathcal{F}_t. \tag{17}$$

To formalize the notion of "available information" at a given time t, we define a filtration. In the context of Brownian motion, the most natural choice is the natural filtration, which represents the history of the process up to time t.

Definition 3 (Filtration). A filtration on $(\Omega, \mathcal{F})$ is an increasing family of sub-σ-algebras $(\mathcal{F}_t)_{t\geq 0}$ such that:

$$\forall s \leq t, \quad \mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}. \tag{18}$$

The natural filtration of a Brownian motion $(B_t)_{t\geq 0}$ is defined by:

$$\mathcal{F}_t \coloneqq \sigma(B_u : 0 \leq u \leq t). \tag{19}$$

When working with a stopping time τ, we need to define the information available up to this *random* time. This is captured by the σ-algebra $\mathcal{F}_\tau$, defined as:

$$\mathcal{F}_\tau \coloneqq \{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t, \, \forall t \geq 0\}. \tag{20}$$

Example: Measurability of the Stopped Process. If τ is a stopping time with respect to the natural filtration $\mathcal{F}_t = \sigma(B_u, 0 \leq u \leq t)$, then the value of the Brownian motion at the random time τ, denoted by B_τ, is $\mathcal{F}_\tau$-measurable.

Significance of Stopping Times for Martingales

Let σ and τ be two stopping times such that $0 \leq \sigma \leq \tau \leq T$ almost surely. If $(M_t)_{t\in[0,T]}$ is a martingale with respect to the filtration $(\mathcal{F}_t)_{t\in[0,T]}$, then the following property holds:

$$\mathbb{E}[M_\tau \mid \mathcal{F}_\sigma] = M_\sigma \quad \text{a.s.} \tag{21}$$

The Markov Property

A fundamental characteristic of Brownian motion is the Markov property. It formalizes the idea that the future evolution of the process, given its current state, is independent of its past history. Formally, we define it as follows:

Definition 4 (Markov Property)*.* Let $(X_t)_{t\geq 0}$ be a Markov process adapted to the filtration $(\mathcal{F}_t)_{t\geq 0}$. For any positive or bounded measurable function f, and for all $s, t \geq 0$, we have:

$$\mathbb{E}[f(X_{s+t}) \mid \mathcal{F}_s] = \mathbb{E}[f(X_{s+t}) \mid X_s]. \tag{22}$$

The Strong Markov Property

While the simple Markov property holds for deterministic times s, the study of barrier options requires a result that remains valid for random times. The Strong Markov Property extends the memoryless nature of a process to stopping times, ensuring that the future depends on the past only through the state reached at the random time τ.

Theorem 2 (Strong Markov Property)**.** *Let $(X_t)_{t\geq 0}$ be a Markov process and let τ be a stopping time with respect to the filtration $(\mathcal{F}_t)_{t\geq 0}$ such that $\mathbb{P}(\tau < \infty) = 1$. Then, for any bounded or positive measurable function f and for any $t \geq 0$:*

$$\mathbb{E}[f(X_{t+\tau}) \mid \mathcal{F}_\tau] = \mathbb{E}[f(X_{t+\tau}) \mid X_\tau]. \tag{23}$$

3.2 Some properties about the Reflection Principle

Having introduced stopping times and the Strong Markov Property, we now possess the necessary tools to analyze the behavior of stochastic processes at random times. In particular, these concepts allow us to study how Brownian motion evolves after reaching a given level.

This framework is essential for understanding path-dependent features, such as barrier crossings, which play a central role in the pricing of barrier options. We are therefore naturally led to examine the Reflection Principle, a key result that exploits both the symmetry of Brownian motion and the Strong Markov Property to describe the distribution of its extrema and hitting times.

First, we demonstrate the reflection principle of Brownian motion, which states that the reflected process is also a Brownian motion. We introduce the process X, representing this reflection, defined by:

$$X_t = B_t \mathbb{1}_{\{t \leq T_a\}} + (2a - B_t)\mathbb{1}_{\{t \geq T_a\}}, \quad t \geq 0$$

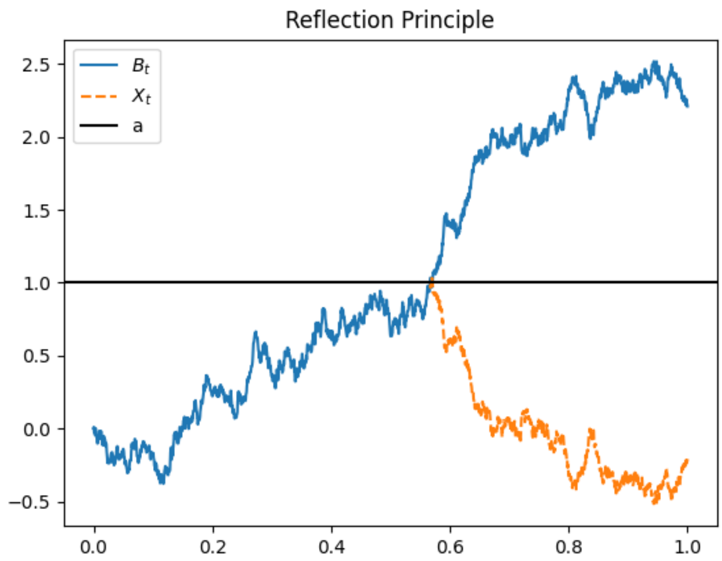

Figure 2: Numerical simulation of X_t.

To this end, we require an intermediate result:

Proposition 1. *Let $(B_t)_{t \geq 0}$ be a Brownian motion equipped with its natural filtration $(\mathcal{F}_t)_{t \geq 0}$ and T_a be the hitting time of a defined by:*

$$T_a := \inf\{t \geq 0 : B_t = a\}$$

Then, for all $x \in \mathbb{R}$ and all $t \geq 0$:

$$\mathbb{P}(2a - B_t \geq x; T_a \leq t) = \mathbb{P}(B_t \geq x; T_a \leq t)$$

Proof. First, observe that:

$$\begin{aligned} 2a - B_t &= a + (B_{T_a} - B_t) \quad (\text{since } B_{T_a} = a) \\ &= a - B_{t-T_a}^{(T_a)} \end{aligned}$$

It follows that:

$$\mathbb{P}(2a - B_t \geq x; T_a \leq t) = \mathbb{P}(-B_{t-T_a}^{(T_a)} \geq x - a; T_a \leq t)$$

By the strong Markov property, T_a is a stopping time, so $B^{(T_a)}_{t-T_a}$ is a Brownian motion independent of $\mathcal{F}_{T_a}$. Consequently, $B^{(T_a)}_{t-T_a}$ is symmetric. It follows that:

$$\begin{aligned}\mathbb{P}(2a - B_t \geq x; T_a \leq t) &= \mathbb{P}(B^{(T_a)}_{t-T_a} \geq x - a; T_a \leq t) \\ &= \mathbb{P}(B_t - B_{T_a} \geq x - B_{T_a}; T_a \leq t) \\ &= \mathbb{P}(B_t \geq x; T_a \leq t) \end{aligned} \tag{24}$$

□

Remark 2. Intuitively, this result was expected given that the process X possesses monotone increments (or deviations) of the same nature as a standard Brownian motion. Since Brownian increments are independent and symmetric, reflecting them after T_a does not alter the statistical law of the trajectory.

We can then conclude:

Proposition 2. *The process X is a Brownian motion.*

Proof. First, by applying the Law of Total Probability, we decompose the distribution of X_t as follows:

$$\begin{aligned}\mathbb{P}(X_t \geq x) &= \mathbb{P}(X_t \geq x, T_a \leq t) + \mathbb{P}(X_t \geq x, T_a > t) \\ &= \mathbb{P}(2a - B_t \geq x, T_a \leq t) + \mathbb{P}(B_t \geq x, T_a > t)\end{aligned}$$

Applying the symmetry result from Proposition 1 to the first term, we obtain:

$$\begin{aligned}&= \mathbb{P}(B_t \geq x, T_a \leq t) + \mathbb{P}(B_t \geq x, T_a > t) \\ &= \mathbb{P}(B_t \geq x)\end{aligned}$$

This establishes the equality in distribution: $X_t \overset{(d)}{=} B_t$.

Next, we verify the second condition of Lévy's Characterization Theorem by computing the quadratic variation $\langle X, X \rangle_t$. To facilitate this, we express the reflected process in the following form:

$$X_t = 2B_{t\wedge T_a} - B_t$$

Using the bilinearity and expansion properties of the quadratic variation operator, we have:

$$\begin{aligned}\langle X, X \rangle_t &= \langle 2B_{\cdot\wedge T_a} - B, 2B_{\cdot\wedge T_a} - B \rangle_t \\ &= 4\langle B_{\cdot\wedge T_a}, B_{\cdot\wedge T_a} \rangle_t - 4\langle B_{\cdot\wedge T_a}, B \rangle_t + \langle B, B \rangle_t\end{aligned}$$

Recall that for a standard Brownian motion, $\langle B, B\rangle_t = t$. Furthermore, for the stopped process, it holds that:

$$\begin{aligned}\langle B_{\cdot\wedge T_a}, B_{\cdot\wedge T_a}\rangle_t &= t \wedge T_a \\ \langle B_{\cdot\wedge T_a}, B\rangle_t &= t \wedge T_a\end{aligned}$$

Substituting these identities into the expression above, we conclude:

$$\begin{aligned}\langle X, X\rangle_t &= 4(t \wedge T_a) - 4(t \wedge T_a) + t \\ &= t\end{aligned}$$

Therefore, by Lévy's Characterization Theorem, X is a standard Brownian motion. □

3.3 Law of the Maximum

We now focus on the maximum reached by our Brownian motion B during its trajectory and attempt to determine the joint law between the Brownian motion and its maximum. We set:

$$S_t = \sup_{s\in[0,t]} B_s \quad \text{where } t > 0$$

and we seek the expression for:

$$F_t(a, x) = \mathbb{P}(S_t \leqslant a \ ; \ B_t \leqslant x)$$

Proposition 3.

$$\mathbb{P}(S_t \geqslant a \ ; \ B_t \leqslant x) = \mathbb{P}(B_t \geqslant 2a - x)$$

Proof. Let us first note that:

$$\{S_t \geqslant a\} = \left\{\sup_{s\in[0,t]} B_s \geqslant a\right\} = \{T_a \leqslant t\}$$

Hence:

$$\begin{aligned}\mathbb{P}(S_t \geqslant a \ ; \ B_t \leqslant x) &= \mathbb{P}(T_a \leqslant t \ ; \ B_t \leqslant x) \\ &= \mathbb{P}(2a - B_t \leqslant x \ ; \ T_a \leqslant t) \\ &= \mathbb{P}(B_t \geqslant 2a - x \ ; \ T_a \leqslant t)\end{aligned}$$

Since $x \leqslant a$, it implies $2a - x \geqslant a$. Thus, on the event $\{B_t \geqslant 2a - x\}$, we necessarily have $B_t \geqslant a$, which ensures that the hitting time T_a has already occurred before t. Therefore, $\{B_t \geqslant 2a - x\} \subset \{T_a \leqslant t\}$.

Finally:

$$\mathbb{P}(S_t \geqslant a \ ; \ B_t \leqslant x) = \mathbb{P}(B_t \geqslant 2a - x)$$

□

Proposition 4. *For all $t \geq 0$, the running maximum S_t has the same distribution as the absolute value of the Brownian motion $|B_t|$.*

Proof. Let $a > 0$. We wish to determine $\mathbb{P}(S_t \geq a)$. By the law of total probability:

$$\mathbb{P}(S_t \geq a) = \mathbb{P}(S_t \geq a, B_t \geq a) + \mathbb{P}(S_t \geq a, B_t < a)$$

If $B_t \geq a$, then the process must have reached a before time t by path continuity (since $B_0 = 0$). Thus, $\{B_t \geq a\} \subset \{T_a \leq t\}$, and the first term simply becomes $\mathbb{P}(B_t \geq a)$.

For the second term, we use the reflection principle (Proposition 1) with $x = a$. We have:

$$\mathbb{P}(S_t \geq a, B_t < a) = \mathbb{P}(2a - B_t > a) = \mathbb{P}(B_t \geqslant 2a - x \ ; \ T_a \leqslant t)$$

By applying the symmetry result demonstrated previously, this probability is equal to $\mathbb{P}(T_a \leq t, B_t > a)$, which is again $\mathbb{P}(B_t > a)$. Consequently:

$$\begin{aligned} \mathbb{P}(S_t \geq a) &= \mathbb{P}(B_t \geq a) + \mathbb{P}(B_t > a) \\ &= 2\mathbb{P}(B_t \geq a) \quad \text{(by the continuity of the law of } B_t\text{)} \end{aligned}$$

Finally, by the symmetry of Brownian motion ($B_t \sim -B_t$), we know that $2\mathbb{P}(B_t \geq a) = \mathbb{P}(B_t \geq a) + \mathbb{P}(B_t \leq -a) = \mathbb{P}(|B_t| \geq a)$. We conclude that for all $a > 0$:

$$\mathbb{P}(S_t \geq a) = \mathbb{P}(|B_t| \geq a)$$

The random variables S_t and $|B_t|$ therefore have the same law. □

Comparison of Processes

Although S_t and $|B_t|$ have the same law for each fixed t, the processes $(S_t)_{t \geq 0}$ and $(|B_t|)_{t \geq 0}$ do not share the same law.

- **Monotonicity argument:** By definition, the paths of $t \mapsto S_t$ are almost surely **increasing** (in the broad sense). In contrast, the paths of $t \mapsto |B_t|$ are not (they oscillate and return to 0 repeatedly).

- **Consequence:** For $s < t$, the joint law (S_s, S_t) is not the same as $(|B_s|, |B_t|)$. For example, $\mathbb{P}(S_t < S_s) = 0$, while $\mathbb{P}(|B_t| < |B_s|) > 0$.

We know that:

$$\begin{aligned}\mathbb{P}(S_t \geqslant a\ ;\ B_t \leqslant x) &= \mathbb{P}(B_t \leqslant x) - \mathbb{P}(S_t \leqslant a\ ;\ B_t \leqslant x)\\ &= \mathbb{P}(B_t \leqslant x) - F_t(a,x)\end{aligned}$$

And thus the joint density is obtained by differentiation. Suppose that $a > 0$ and $x < a$:

$$\begin{aligned}f_t(a,x) = \frac{\partial^2 F_t}{\partial a \partial x}(a,x) &= 0 - \frac{\partial^2}{\partial a \partial x}\mathbb{P}(S_t \geqslant a\ ;\ B_t \leqslant x)\\ &= -\frac{\partial^2}{\partial a \partial x}\mathbb{P}(B_t \geqslant 2a - x) \quad \text{by Proposition 3}\\ &= -\frac{\partial^2}{\partial a \partial x}\left(1 - F_{B_t}(2a-x)\right)\\ &= \frac{\partial^2}{\partial a \partial x}\big(F_{B_t}(2a-x)\big)\\ &= \frac{\partial}{\partial x}\Big(2 f_{B_t}(2a-x)\Big)\\ &= \frac{\partial}{\partial x}\left(\frac{2}{\sqrt{2\pi t}}\mathrm{e}^{-\frac{(2a-x)^2}{2t}}\right)\\ &= \frac{2}{\sqrt{2\pi t}} \cdot \left(\frac{2a-x}{t}\right) \cdot \mathrm{e}^{-\frac{(2a-x)^2}{2t}}\\ &= \frac{2(2a-x)}{\sqrt{2\pi t^3}}\mathrm{e}^{-\frac{(2a-x)^2}{2t}}\end{aligned}$$

If $a < 0$:

$$\mathbb{P}(S_t \leqslant a; B_t \leqslant x) = 0$$

since $B_0 = 0$ implies $S_t \geqslant 0$.

If $x > a$:

$$\mathbb{P}(B_t \leqslant x; S_t \leqslant a) = \mathbb{P}(S_t \leqslant a)$$

Thus:

$$f_t(a,x) = \frac{\partial^2}{\partial a \partial x}\mathbb{P}(S_t \leqslant a) = 0$$

Finally,

$$f_t(a,x) = \frac{2(2a-x)}{\sqrt{2\pi t^3}} \exp\left(-\frac{(2a-x)^2}{2t}\right) \mathbb{1}_{[a,+\infty[}(x)\mathbb{1}_{[0,+\infty[}(a) \tag{25}$$

We aim to determine the probability density of the hitting time T_a, denoted $f_{T_a}(t)$. We first express its distribution function $\mathbb{P}(T_a \leq t)$.

Proposition 5. *The density of the first passage time T_a for $a > 0$ is given by:*

$$f_{T_a}(t) = \frac{a}{\sqrt{2\pi t^3}}\mathrm{e}^{-\frac{a^2}{2t}}\mathbb{1}_{\{t>0\}}$$

Proof. Let $a > 0$. Recall the fundamental relation established previously between the current maximum S_t and the hitting time T_a:

$$\{T_a \leq t\} = \{S_t \geq a\}$$

From Proposition 4, we know that:

$$\mathbb{P}(S_t \geq a) = 2\mathbb{P}(B_t \geq a)$$

We derive the distribution function of T_a using the normal law $\mathcal{N}(0, t)$ of the Brownian motion:

$$F_{T_a}(t) = \mathbb{P}(T_a \leq t) = 2\int_a^{+\infty} \frac{1}{\sqrt{2\pi t}} \mathrm{e}^{-\frac{u^2}{2t}} \,\mathrm{d}u$$

To obtain the density, we differentiate this expression with respect to t using the Leibniz rule. Let us use the change of variable $v = \frac{u}{\sqrt{t}}$, hence $\mathrm{d}u = \sqrt{t}\,\mathrm{d}v$:

$$F_{T_a}(t) = 2\int_{\frac{a}{\sqrt{t}}}^{+\infty} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{v^2}{2}} \,\mathrm{d}v$$

The derivative of this function with respect to t is:

$$\begin{aligned} f_{T_a}(t) = \frac{\mathrm{d}}{\mathrm{d}t} F_{T_a}(t) &= 2 \cdot \left(-\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{a}{\sqrt{t}}\right)\right) \cdot \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{(a/\sqrt{t})^2}{2}} \\ &= 2 \cdot \left(\frac{a}{2t\sqrt{t}}\right) \cdot \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{a^2}{2t}} \\ &= \frac{a}{t\sqrt{2\pi t}} \mathrm{e}^{-\frac{a^2}{2t}} \end{aligned}$$

Simplifying the terms, we finally obtain:

$$f_{T_a}(t) = \frac{a}{\sqrt{2\pi t^3}} \exp\left(-\frac{a^2}{2t}\right) \mathbb{1}_{\{t>0\}}$$

□

4 Drifted Brownian Motion

Building upon the results established in the previous chapter, we now have a solid grasp of joint distributions and hitting times within a standard Brownian framework. While these findings are foundational, they rely heavily on the spatial symmetry of the process.

In financial modeling, however, asset prices typically exhibit a trend or drift, which breaks this symmetry. To bridge the gap toward realistic option pricing, we must extend our analysis to drifted Brownian motion, utilizing change-of-measure techniques to characterize its extrema.

4.1 Brownian Motion with Drift

In this section, we apply Girsanov's theorem to determine the density function for the maximum of a drifted Brownian motion $\tilde{B}_t = B_t + \mu t$, where $\mu \in \mathbb{R}$. This result is a prerequisite for the analytical valuation of barrier-dependent derivatives. Let the running maximum be defined as:

$$\tilde{X}_T = \max_{t\in[0,T]} \tilde{B}_t = \max_{t\in[0,T]} (B_t + \mu t)$$

Recall that under the risk-neutral measure $\mathbb{Q}$, the process $\tilde{B}_t$ behaves as a standard Brownian motion. The Radon-Nikodym derivative defining this change of measure is:

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = e^{-\mu B_T - \mu^2 T/2}$$

By leveraging the joint density of a standard Brownian motion and its maximum, we can express the probability under $\mathbb{P}$ for any $0 \le b \le a$ as follows:

$$\begin{aligned}
\mathbb{P}(\tilde{X}_T \le a \ \&\ \tilde{B}_T \le b) &= \mathbb{E}\left[\mathbf{1}_{\{\tilde{X}_T \le a \ \&\ \tilde{B}_T \le b\}}\right] \\
&= \mathbb{E}^{\mathbb{Q}}\left[e^{\mu \tilde{B}_T - \mu^2 T/2}\mathbf{1}_{\{\tilde{X}_T \le a \ \&\ \tilde{B}_T \le b\}}\right] \\
&= \sqrt{\frac{2}{\pi T}} \int_0^a \int_{-\infty}^b \mathbf{1}_{[0,x]}(y) e^{\mu y - \mu^2 T/2} \frac{(2x-y)}{T} e^{-(2x-y)^2/(2T)} dx dy
\end{aligned}$$

Differentiating this expression with respect to a and b yields the joint density:

$$f_{\tilde{X}_T,\tilde{B}_T}(a,b) = \mathbf{1}_{\{a \ge b \wedge 0\}} \frac{1}{T}\sqrt{\frac{2}{\pi T}}(2a-b)e^{\mu b - \frac{(2a-b)^2}{2T} - \frac{\mu^2 T}{2}}$$

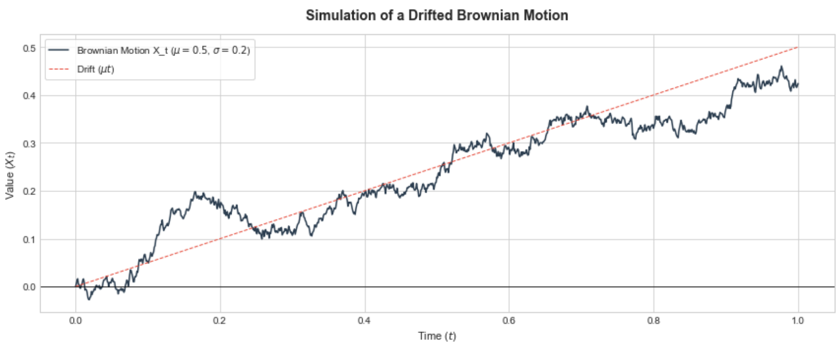

Figure 3: Numerical simulation of $\tilde{X}_T$.

4.2 Barrier Options

General Case

This section focuses on the analytical pricing of barrier options. Unlike European options, which are sensitive only to the terminal price, barrier options are path-dependent exotic derivatives. Their payoff is conditional upon the underlying asset's price reaching or avoiding a specific threshold, known as the barrier, during the contract's life.

The fair value of such a claim at time t is represented by the discounted conditional expectation:

$$e^{-r(T-t)}\mathbb{E}\left[\phi(\tilde{X}_T, \tilde{B}_T) \mid \mathcal{F}_t\right]$$

where $\phi(\cdot,\cdot)$ denotes the payoff function. At the inception of the contract ($t = 0$), this value is computed by integrating over the joint density derived in the previous section:

$$e^{-rT}\int_{-\infty}^{\infty}\int_{y\vee 0}^{\infty}\phi(x,y)f_{\tilde{X}_T,\tilde{B}_T}(x,y)\,dxdy.$$

In our framework, the payoff depends on the spot price S_T and the maximum M_T. Under the drifted motion, these variables are linked to $\tilde{B}_T$ and $\tilde{X}_T$ as follows:

$$\begin{aligned}
C &= \phi(S_T, M_T)\\
&= \phi(S_0 e^{\sigma B_T - \sigma^2 T/2 + rT}, \max_{t\in[0,T]} S_t)\\
&= \phi(S_0 e^{\sigma \tilde{B}_T}, S_0 \max_{t\in[0,T]} e^{\sigma B_t - \sigma^2 t/2 + rt})\\
&= \phi(S_0 e^{\sigma \tilde{B}_T}, S_0 \max_{t\in[0,T]} e^{\sigma \tilde{B}_t})\\
&= \phi(S_0 e^{\sigma \tilde{B}_T}, S_0 e^{\sigma \max_{t\in[0,T]} \tilde{B}_t})\\
&= \phi(S_0 e^{\sigma \tilde{B}_T}, S_0 e^{\sigma \tilde{X}_T}).
\end{aligned}$$

with the drift parameter $\mu = (r - \sigma^2/2)/\sigma$.

Up-and-out Barrier Call Option

An Up-and-out call option is a specific type of barrier option that provides a standard call payoff unless the asset price hits a high barrier B, in which case the option is "knocked out" and expires worthless. Its mathematical payoff at maturity T is:

$$C = (S_T - K)^+ \mathbf{1}_{\{M_T \leq B\}} \tag{26}$$

where $M_T = \sup_{0\leq t\leq T} S_t$ is the running maximum. We assume $K < B$ to ensure the option is financially viable, as a strike price higher than the barrier would result in a null payoff in every possible scenario.

Proposition 6. *The no-arbitrage price of an up-and-out call option at time t is given by:*

$$
\begin{aligned}
V_t &= e^{-r(T-t)}\mathbb{E}\left[C \mid \mathcal{F}_t\right] &(27)\\
&= S_t\mathbf{1}_{\{M_t\le B\}}\left[\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{K}\right)\right)-\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{B}\right)\right)\right.\\
&\quad\left.-\left(\frac{B}{S_t}\right)^{1+2r/\sigma^2}\left(\Phi\left(\delta_+^{T-t}\left(\frac{B^2}{KS_t}\right)\right)-\Phi\left(\delta_+^{T-t}\left(\frac{B}{S_t}\right)\right)\right)\right]\\
&\quad-e^{-r(T-t)}K\mathbf{1}_{\{M_t\le B\}}\left[\Phi\left(\delta_-^{T-t}\left(\frac{S_t}{K}\right)\right)-\Phi\left(\delta_-^{T-t}\left(\frac{S_t}{B}\right)\right)\right.\\
&\quad\left.-\left(\frac{S_t}{B}\right)^{1-2r/\sigma^2}\left(\Phi\left(\delta_-^{T-t}\left(\frac{B^2}{KS_t}\right)\right)-\Phi\left(\delta_-^{T-t}\left(\frac{B}{S_t}\right)\right)\right)\right]. &(28)
\end{aligned}
$$

Proof. To facilitate the derivation, we first establish three key components. First, the evaluation of the path-dependent integral:

$$
\begin{aligned}
\int_a^b (2v-u)e^{\frac{2v(u-v)}{\tau}}\,dv &= \left[-\frac{\tau}{2}e^{\frac{2v(u-v)}{\tau}}\right]_{v=a}^{v=b} \\
&= \frac{\tau}{2}\left(e^{\frac{2a(u-a)}{\tau}} - e^{\frac{2b(u-b)}{\tau}}\right) \\
&= \frac{\tau}{2}\left(1 - e^{\frac{2b(u-b)}{\tau}}\right).
\end{aligned} \tag{29}
$$

Second, the transformation of Gaussian integrals into the standard normal CDF, $\Phi(x)$:

$$
\frac{1}{\sqrt{2\pi\tau}}\int_c^b e^{\gamma u - \frac{u^2}{2\tau}}\,du = e^{\frac{\gamma^2\tau}{2}}\left(\Phi\left(\frac{-c+\gamma\tau}{\sqrt{\tau}}\right) - \Phi\left(\frac{-b+\gamma\tau}{\sqrt{\tau}}\right)\right). \tag{30}
$$

Finally, we use the standard Black-Scholes variables $\delta_{\pm}^{\tau}(s)$.

$$
\forall s>0,\ \delta_{\pm}^{\tau}(s) = \frac{1}{\sigma\sqrt{\tau}}\left(\log s + \left(r \pm \frac{1}{2}\sigma^2\right)\tau\right). \tag{31}
$$

Everything is now wet up so that we can calculate the price of an up-and-out call option. We begin by :

$$
\begin{aligned}
e^{-r(T-t)}\mathbb{E}\left[C \mid \mathcal{F}_t\right] &= e^{-r(T-t)}\mathbb{E}\left[(S_T-K)^+\mathbf{1}_{\{M_T\le B\}} \mid \mathcal{F}_t\right] \\
&= e^{-r(T-t)}\mathbb{E}\left[(S_T-K)^+\mathbf{1}_{\{M_t\le B\}}\mathbf{1}_{\{\max_{t\le r\le T} S_r\le B\}} \mid \mathcal{F}_t\right] \\
&= e^{-r(T-t)}\mathbf{1}_{\{M_t\le B\}}\mathbb{E}\left[(S_T-K)^+\mathbf{1}_{\{\max_{t\le r\le T} S_r\le B\}} \mid \mathcal{F}_t\right] \\
&= e^{-r(T-t)}\mathbf{1}_{\{M_t\le B\}}\mathbb{E}\left[\left(x\frac{S_T}{S_t}-K\right)^+\mathbf{1}_{\{x\max_{t\le r\le T} S_r/S_t\le B\}}\right]_{x=S_t} \\
&= e^{-r(T-t)}\mathbf{1}_{\{M_t\le B\}}\mathbb{E}\left[\left(x\frac{S_{T-t}}{S_0}-K\right)^+\mathbf{1}_{\{x\max_{0\le r\le T-t} S_r/S_0\le B\}}\right]_{x=S_t}.
\end{aligned}
$$

Using the change of variables $\tau = T-t$, $\frac{S_{T-t}}{S_0} = e^{\sigma\tilde{B}_\tau}$ and $\frac{\max S_r}{S_0} = e^{\sigma\tilde{X}_\tau}$, the expression becomes:

$$
e^{-r\tau}\mathbf{1}_{\{M_t\le B\}}\mathbb{E}\left[\left(xe^{\sigma\tilde{B}_\tau}-K\right)^+\mathbf{1}_{\{\tilde{X}_\tau\le\frac{1}{\sigma}\ln(B/x)\}}\right]_{x=S_t}
$$

The indicator $\mathbf{1}_{\{M_t\le B\}}$ stays outside the integral because it is measurable with respect to the filtration $\mathcal{F}_t$ (the past is known). The change of variable only applies to the future stochastic part. By introducing the joint density $f(u,v)$ of the couple $(\tilde{B}_\tau, \tilde{X}_\tau)$, we transform this expectation into the following double integral:

$$
e^{-r\tau}\mathbf{1}_{\{M_t\le B\}}\int_{-\infty}^{\frac{1}{\sigma}\ln(B/x)}\int_{u\vee 0}^{\frac{1}{\sigma}\ln(B/x)}(xe^{\sigma u}-K)^+ f(u,v)\,dv\,du\Bigg|_{x=S_t}
$$

$$
\begin{aligned}
&e^{-r\tau}\mathbf{1}_{\{M_t\le B\}}\mathbb{E}\left[\left(xe^{\sigma\tilde{B}_\tau}-K\right)^+\mathbf{1}_{\{xe^{\sigma\tilde{X}_\tau}\le B\}}\right]_{x=S_t}\\
&=e^{-r\tau}\mathbf{1}_{\{M_t\le B\}}\left[\int_{-\infty}^{\infty}\int_{u\vee 0}^{\infty}(xe^{\sigma u}-K)^+\mathbf{1}_{\{xe^{\sigma v}\le B\}}f(u,v)\,dv\,du\right]_{x=S_t}\\
&=\frac{1}{\tau}\sqrt{\frac{2}{\pi\tau}}e^{-r\tau}\mathbf{1}_{\{M_t\le B\}}\left[\int_{-\infty}^{\sigma^{-1}\log(B/x)}\int_{u\vee 0}^{\infty}(xe^{\sigma u}-K)^+\mathbf{1}_{\{xe^{\sigma v}\le B\}}\right.\\
&\qquad\qquad\left.\times(2v-u)e^{-\frac{\mu^2\tau}{2}+\mu u-\frac{(2v-u)^2}{2\tau}}\,dv\,du\right]_{x=S_t}\\
&=\frac{1}{\tau}e^{-r\tau}\sqrt{\frac{2}{\pi\tau}}\mathbf{1}_{\{M_t\le B\}}\left[\int_{0}^{\sigma^{-1}\log(B/x)}\int_{u}^{\infty}(xe^{\sigma u}-K)^+\mathbf{1}_{\{v\le\sigma^{-1}\log(B/x)\}}(2v-u)e^{\cdots}\,dv\,du\right.\\
&\qquad\left.+\int_{-\infty}^{0}\int_{0}^{\infty}(xe^{\sigma u}-K)^+\mathbf{1}_{\{v\le\sigma^{-1}\log(B/x)\}}(2v-u)e^{\cdots}\,dv\,du\right]_{x=S_t}\\
&=\frac{1}{\tau}e^{-r\tau}\sqrt{\frac{2}{\pi\tau}}\mathbf{1}_{\{M_t\le B\}}\left[\int_{0}^{\sigma^{-1}\log(B/x)}\int_{u}^{\sigma^{-1}\log(B/x)}(xe^{\sigma u}-K)^+(2v-u)e^{-\frac{\mu^2\tau}{2}+\mu u-\frac{(2v-u)^2}{2\tau}}\,dv\,du\right.\\
&\qquad\left.+\int_{-\infty}^{0}\int_{0}^{\sigma^{-1}\log(B/x)}(xe^{\sigma u}-K)^+(2v-u)e^{-\frac{\mu^2\tau}{2}+\mu u-\frac{(2v-u)^2}{2\tau}}\,dv\,du\right]_{x=S_t}\\
&=\frac{1}{\tau}e^{-r\tau}\sqrt{\frac{2}{\pi\tau}}\mathbf{1}_{\{M_t\le B\}}\left[\int_{\sigma^{-1}\log(K/x)}^{\sigma^{-1}\log(B/x)}\int_{u\vee 0}^{\sigma^{-1}\log(B/x)}(xe^{\sigma u}-K)(2v-u)e^{-\frac{\mu^2\tau}{2}+\mu u-\frac{(2v-u)^2}{2\tau}}\,dv\,du\right]_{x=S_t}\\
&=\frac{1}{\tau}e^{-r\tau-\frac{\mu^2\tau}{2}}\sqrt{\frac{2}{\pi\tau}}\mathbf{1}_{\{M_t\le B\}}\left[\int_{\sigma^{-1}\log(K/x)}^{\sigma^{-1}\log(B/x)}(xe^{\sigma u}-K)e^{\mu u-\frac{u^2}{2\tau}}\right.\\
&\qquad\left.\times\int_{u\vee 0}^{\sigma^{-1}\log(B/x)}(2v-u)e^{\frac{2v(u-v)}{\tau}}\,dv\,du\right]_{x=S_t}
\end{aligned}
$$

Now we can use the equation (29), by setting $a=u\vee 0$ and $b=\sigma^{-1}\log(B/x)$. To simplify even more we set $c=\sigma^{-1}\log(K/x)$ and we get :

$$
\begin{aligned}
e^{-r\tau}\mathbb{E}[C]=\mathbf{1}_{\{M_t\le B\}}&\left[xe^{-\tau(r+\frac{\mu^2}{2})}\frac{1}{\sqrt{2\pi\tau}}\int_c^b e^{u(\sigma+\mu)-\frac{u^2}{2\tau}}\,du\right.\\
&-xe^{-\tau(r+\frac{\mu^2}{2})-\frac{2b^2}{\tau}}\frac{1}{\sqrt{2\pi\tau}}\int_c^b e^{u(\sigma+\mu+\frac{2b}{\tau})-\frac{u^2}{2\tau}}\,du\\
&-Ke^{-\tau(r+\frac{\mu^2}{2})}\frac{1}{\sqrt{2\pi\tau}}\int_c^b e^{\mu u-\frac{u^2}{2\tau}}\,du\\
&\left.+Ke^{-\tau(r+\frac{\mu^2}{2})-\frac{2b^2}{\tau}}\frac{1}{\sqrt{2\pi\tau}}\int_c^b e^{u(\mu+\frac{2b}{\tau})-\frac{u^2}{2\tau}}\,du\right]_{x=S_t}
\end{aligned}
$$

Thanks to the equation (30) we find :

$$
\begin{aligned}
e^{-r\tau}\mathbb{E}[C] = \mathbf{1}_{\{M_t\le B\}}\Bigg[& x e^{-\tau(r+\frac{\mu^2}{2})+\frac{(\sigma+\mu)^2\tau}{2}}\left(\Phi\left(\frac{-c+(\sigma+\mu)\tau}{\sqrt{\tau}}\right)-\Phi\left(\frac{-b+(\sigma+\mu)\tau}{\sqrt{\tau}}\right)\right)\\
& - x e^{-\tau(r+\frac{\mu^2}{2})-\frac{2b^2}{\tau}+\frac{(\sigma+\mu+\frac{2b}{\tau})^2\tau}{2}}\left(\Phi\left(\frac{-c+(\sigma+\mu+\frac{2b}{\tau})\tau}{\sqrt{\tau}}\right)-\Phi\left(\frac{-b+(\sigma+\mu+\frac{2b}{\tau})\tau}{\sqrt{\tau}}\right)\right)\\
& - K e^{-\tau(r+\frac{\mu^2}{2})+\frac{\mu^2\tau}{2}}\left(\Phi\left(\frac{-c+\mu\tau}{\sqrt{\tau}}\right)-\Phi\left(\frac{-b+\mu\tau}{\sqrt{\tau}}\right)\right)\\
& + K e^{-\tau(r+\frac{\mu^2}{2})-\frac{2b^2}{\tau}+\frac{(\mu+\frac{2b}{\tau})^2\tau}{2}}\left(\Phi\left(\frac{-c+(\mu+\frac{2b}{\tau})\tau}{\sqrt{\tau}}\right)-\Phi\left(\frac{-b+(\mu+\frac{2b}{\tau})\tau}{\sqrt{\tau}}\right)\right)\Bigg]_{x=S_t}
\end{aligned}
$$

Using the equation (31) and those relations:

$$
\begin{aligned}
-\tau(r+\mu^2/2)-2b^2/\tau+(\sigma+\mu+2b/\tau)^2\tau/2 &= (1+2r/\sigma^2)\log(B/x),\\
-\tau(r+\mu^2/2)-2b^2/\tau+(\mu+2b/\tau)^2\tau/2 &= -r\tau+(-1+2r/\sigma^2)\log(B/x),
\end{aligned}
$$

we can reduce the final expression to :

$$
\begin{aligned}
e^{-r\tau}\mathbb{E}[C] = \mathbf{1}_{\{M_t\le B\}}\Bigg[&\\
& x\left(\Phi\left(\delta_+^\tau\left(\frac{x}{K}\right)\right)-\Phi\left(\delta_+^\tau\left(\frac{x}{B}\right)\right)\right)\\
& - x\left(\frac{B}{x}\right)^{1+2r/\sigma^2}\left(\Phi\left(\delta_+^\tau\left(\frac{B^2}{Kx}\right)\right)-\Phi\left(\delta_+^\tau\left(\frac{B}{x}\right)\right)\right)\\
& - e^{-r\tau}K\left(\Phi\left(\delta_-^\tau\left(\frac{x}{K}\right)\right)-\Phi\left(\delta_-^\tau\left(\frac{x}{B}\right)\right)\right)\\
& + e^{-r\tau}K\left(\frac{x}{B}\right)^{1-2r/\sigma^2}\left(\Phi\left(\delta_-^\tau\left(\frac{B^2}{Kx}\right)\right)-\Phi\left(\delta_-^\tau\left(\frac{B}{x}\right)\right)\right)\Bigg]_{x=S_t}
\end{aligned}
$$

The integration yields a deterministic expression based on the current underlying value x and the remaining time $\tau = T - t$. By applying the substitution theorem for conditional expectations and the Markov property of Geometric Brownian Motion, we evaluate this result at $x = S_t$.

$$
\begin{aligned}
e^{-r(T-t)}\mathbb{E}\left[C \mid \mathcal{F}_t\right] = S_t\mathbf{1}_{\{M_t\le B\}}\Bigg[&\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{K}\right)\right) - \Phi\left(\delta_+^{T-t}\left(\frac{S_t}{B}\right)\right) \\
&- \left(\frac{B}{S_t}\right)^{1+2r/\sigma^2}\left(\Phi\left(\delta_+^{T-t}\left(\frac{B^2}{KS_t}\right)\right) - \Phi\left(\delta_+^{T-t}\left(\frac{B}{S_t}\right)\right)\right)\Bigg] \\
- e^{-r(T-t)}K\mathbf{1}_{\{M_t\le B\}}\Bigg[&\Phi\left(\delta_-^{T-t}\left(\frac{S_t}{K}\right)\right) - \Phi\left(\delta_-^{T-t}\left(\frac{S_t}{B}\right)\right) \\
&- \left(\frac{S_t}{B}\right)^{1-2r/\sigma^2}\left(\Phi\left(\delta_-^{T-t}\left(\frac{B^2}{KS_t}\right)\right) - \Phi\left(\delta_-^{T-t}\left(\frac{B}{S_t}\right)\right)\right)\Bigg].
\end{aligned}
$$

The final analytical formula provides the exact price for an Up-and-Out call (equation (28)). Financially, this can be interpreted as the price of a standard European call adjusted by a "knock-out discount" that accounts for the risk of deactivation. This closed-form solution serves as a vital benchmark for validating the accuracy of the numerical algorithms, such as Monte-Carlo simulations, that we will implement in the subsequent stages of this project.

The Partial Differential Equations (PDE) Approach

While the probabilistic approach provided a robust path-wise valuation of the Up-and-Out Call, we now shift our perspective toward a local, dynamic framework. The motivation for this transition is twofold: first, to establish a hedging strategy (the Delta) that ensures market consistency, and second, to verify that our global solution satisfies the fundamental equilibrium equations of the Black-Scholes model.

In this context, the option price at time t can be expressed as a function $g(t, S_t, M_t)$. However, a key observation in the barrier framework is that as long as the underlying price S_t remains strictly below the historical maximum M_t and the barrier B, the price does not dynamically depend on M_t. Consequently, we can observe that:

$$\frac{\partial g}{\partial y}(t, x, y) = 0, \qquad \text{for } 0 < y < B.$$

This allows us to simplify our study to a function $g(t, x)$, focusing on the price dynamics within the "safe" zone.

Proposition 7. *Consider a self-financing portfolio $(\eta_t, \xi_t)_{t\in\mathbb{R}_+}$ (as defined in Section 2.2) such that its value matches the option price $V_t = g(t, S_t)$. On the domain $t \in [0, T]$ and $x \in (0, B)$, the function $g(t, x)$ must satisfy the standard Black-Scholes PDE:*

$$rg(t, x) = \frac{\partial g}{\partial t}(t, x) + rx\frac{\partial g}{\partial x}(t, x) + \frac{1}{2}x^2\sigma^2\frac{\partial^2 g}{\partial x^2}(t, x).$$

Furthermore, the delta-hedging strategy is given by $\xi_t = \frac{\partial g}{\partial x}(t, S_t)$, provided that the barrier has not been breached ($M_t < B$).

To uniquely characterize the price of the Up-and-Out Call, we must supplement this PDE with boundary conditions that reflect the "knock-out" feature and the final payoff.

Proposition 8. *The pricing function $g(t,x)$ for an Up-and-Out Call with strike K and barrier B is the unique solution to the following system on $[0,T] \times [0,B]$:*

$$\begin{cases} rg(t,x) = \frac{\partial g}{\partial t}(t,x) + rx\frac{\partial g}{\partial x}(t,x) + \frac{1}{2}x^2\sigma^2\frac{\partial^2 g}{\partial x^2}(t,x) \\ g(t,B) = 0 \qquad \textit{(Knock-out condition)} \\ g(T,x) = (x-K)^+\mathbf{1}_{\{x<B\}} \qquad \textit{(Terminal payoff)} \\ g(t,0) = 0 \qquad \textit{(Boundary at zero)} \end{cases}$$

The closed-form analytical solution for this PDE was previously demonstrated via the probabilistic approach. Using the variable x to denote the underlying price, the solution is written:

$$\begin{aligned} g(t,x) = x &\left(\Phi\left(\delta_+^{T-t}\left(\frac{x}{K}\right)\right) - \Phi\left(\delta_+^{T-t}\left(\frac{x}{B}\right)\right)\right) \\ &- x\left(\frac{x}{B}\right)^{1-\frac{2r}{\sigma^2}}\left(\Phi\left(\delta_+^{T-t}\left(\frac{B^2}{Kx}\right)\right) - \Phi\left(\delta_+^{T-t}\left(\frac{B}{x}\right)\right)\right) \\ &- Ke^{-r(T-t)}\left(\Phi\left(\delta_-^{T-t}\left(\frac{x}{K}\right)\right) - \Phi\left(\delta_-^{T-t}\left(\frac{x}{B}\right)\right)\right) \\ &+ Ke^{-r(T-t)}\left(\frac{x}{B}\right)^{1-\frac{2r}{\sigma^2}}\left(\Phi\left(\delta_-^{T-t}\left(\frac{B^2}{Kx}\right)\right) - \Phi\left(\delta_-^{T-t}\left(\frac{B}{x}\right)\right)\right), \end{aligned}$$

for $0 < x \leq B$ and $0 \leq t \leq T$.

It is essential to point out that we did not need to solve this Partial Differential Equation analytically from scratch (for example via the method of images). Indeed, by virtue of the Feynman-Kac theorem, the unique solution of this Black-Scholes PDE, subject to our terminal and barrier conditions, exactly matches the conditional expectation calculated in the previous section under the risk-neutral probability. Thus, both approaches: the probabilistic approach (global) and the PDE approach (local) strictly converge to the same closed formula, mutually validating our results.

Verification of the Boundary Conditions

To validate our previous analytical result, we verify that the closed-form solution satisfies these constraints. We focus specifically on the deactivation condition at the barrier ($x = B$).

When $x = B$, our analytical formula (derived in Section 4.2.2) yields:

$$\begin{aligned}
g(t, B) = B &\left[\Phi\left(\delta_+^{T-t}\left(\frac{B}{K}\right)\right) - \Phi\left(\delta_+^{T-t}(1)\right)\right] \\
&- e^{-r(T-t)} K \left[\Phi\left(\delta_-^{T-t}\left(\frac{B}{K}\right)\right) - \Phi\left(\delta_-^{T-t}(1)\right)\right] \\
&- B \left[\Phi\left(\delta_+^{T-t}\left(\frac{B}{K}\right)\right) - \Phi\left(\delta_+^{T-t}(1)\right)\right] \\
&+ e^{-r(T-t)} K \left[\Phi\left(\delta_-^{T-t}\left(\frac{B}{K}\right)\right) - \Phi\left(\delta_-^{T-t}(1)\right)\right] \\
= 0.&
\end{aligned}$$

The cancellation of these terms demonstrates that the model correctly anticipates the immediate loss of value upon contact with B.

The remaining conditions are also verified as follows:

- At maturity ($t = T$): The terms $\delta_\pm$ diverge to $\pm\infty$ depending on the ratio x/K and x/B, effectively reducing the formula to the payoff $(x - K)^+ \mathbf{1}_{\{x<B\}}$.
- At $x = 0$: The price naturally converges to zero, consistent with the absorption of the underlying process.
- For $x > B$: By definition, the function is extended to $g(t, x) = 0$, representing the post-breach state.

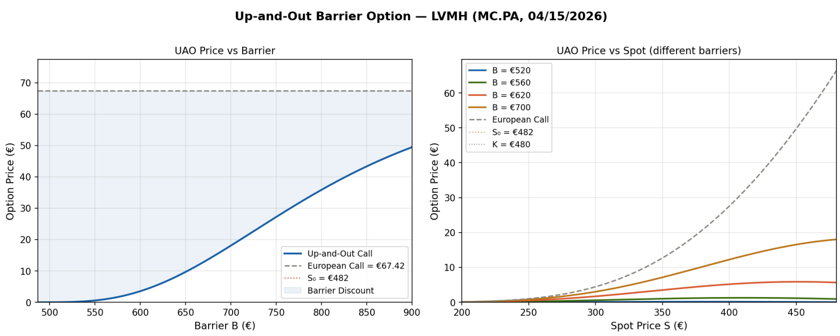

Figure 4: UAO Barrier Option - LVMH.

We simulated the pricing of an Up-and-Out (UAO) barrier option on LVMH stock using actual data from April 15, 2026 — spot price $S_0 = 481.75$, historical volatility $\sigma = 32\%$, ECB rate $r = 2.5\%$ — with a maturity $T = 1$ year and an ATM strike price $K = 480$.

The first graph shows that the closer the barrier B is to the spot price ($S_0 = 481.75$), the more the UAO option is discounted compared to a standard call — logically, the knock-out risk is high. When B exceeds 750, the two prices converge because the barrier becomes virtually unattainable.

The second graph illustrates the fundamental paradox of the UAO: unlike a classic call option whose price rises monotonically with the spot price, the UAO price collapses when S approaches B — the option loses value even if the stock price rises, because the risk of cancellation becomes dominant. The higher the barrier ($B = 700$), the more the curve resembles that of a classic call option.

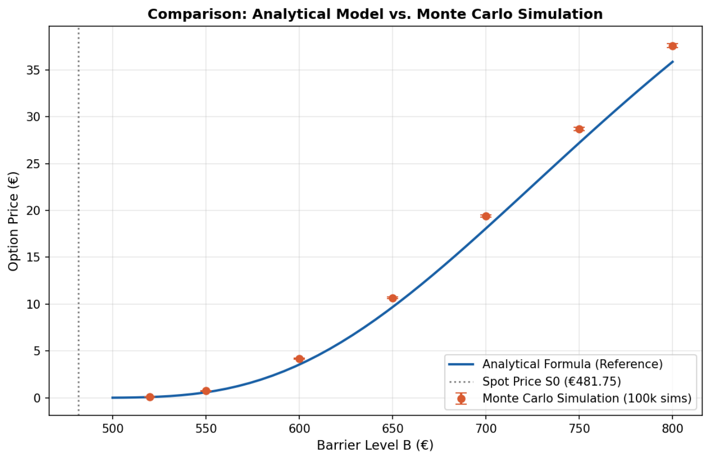

Figure 5: Monte Carlo vs Analytical UAO.

We also wanted to simulate the price thanks to the Monte Carlo simulation we can see from the graph that the results are practically the same except that MC often slightly overestimates the price for discrete barrier options because continuous paths might cross the barrier between daily observation points.

Consistency and Financial Implications

While the probabilistic approach gives us the "fair value" of a barrier option, the PDE-based approach is preferred by traders because it provides essential information known as the Greeks. The most important of these is the Delta (Δ), defined as the sensitivity of the option's price to changes in the underlying asset:

$$\Delta = \frac{\partial g}{\partial x}(t, x) \tag{32}$$

In the real world, the Delta acts as a hedging guide. If a bank sells a UAO (Up-and-Out) call with a Delta of, for example, 0.6, the trader must buy exactly 0.6 units of the underlying stock to hedge the bank against price fluctuations. Thanks to the PDE, the trader can adjust this position in real-time as the market moves.

Indeed, the boundary condition $g(t, B) = 0$ is not just a mathematical constraint; it serves as a risk management signal. For a standard call, the value of the Delta will always be positive. However, for a UAO option, the Delta changes drastically as the price approaches the barrier B:

- Away from the barrier: The option behaves like a standard call, and the trader maintains a long position in the stock.
- Near the barrier: The "knock-out" risk becomes extremely high. If the price touches B, the option becomes worthless. To anticipate this, the Delta can drop sharply or even become negative.

This leads to a phenomenon known as Pin Risk. When the asset price is "pinned" near the barrier, a tiny market move can decide whether the option lives or dies. This forces the trader to buy or sell massive amounts of stock very quickly to remain hedged. By integrating the barrier directly into our PDE, we ensure that the hedging strategy remains consistent, even when facing this "financial cliff."

Down-and-out Barrier Call Option

While the previous derivation focused on the Up-and-Out case, where the asset price is constrained by an upper boundary, a similar logic applies to Down-and-Out options. By leveraging the same risk-neutral valuation framework and the Feynman-Kac theorem, we can establish the pricing system for a barrier B located below the initial spot price. The mathematical treatment remains analogous: the "knock-out" event is now triggered if the asset price falls to or below B, shifting the domain of the solution and the corresponding boundary conditions.

Proposition 9. *The pricing function $h(t,x)$ for a Down-and-Out Call with strike K and barrier B (where $B < S_0$ and typically $B < K$) is the unique solution to the following system on $[0,T] \times [B, +\infty[$:*

$$\begin{cases} rh(t,x) = \frac{\partial h}{\partial t}(t,x) + rx\frac{\partial h}{\partial x}(t,x) + \frac{1}{2}x^2\sigma^2\frac{\partial^2 h}{\partial x^2}(t,x) \\ h(t,B) = 0 \qquad \textit{(Knock-out condition)} \\ h(T,x) = (x-K)^+\mathbf{1}_{\{x>B\}} \qquad \textit{(Terminal payoff)} \\ \lim_{x\to\infty}\left[h(t,x) - (x - Ke^{-r(T-t)})\right] = 0 \qquad \textit{(Asymptotic behavior)} \end{cases}$$

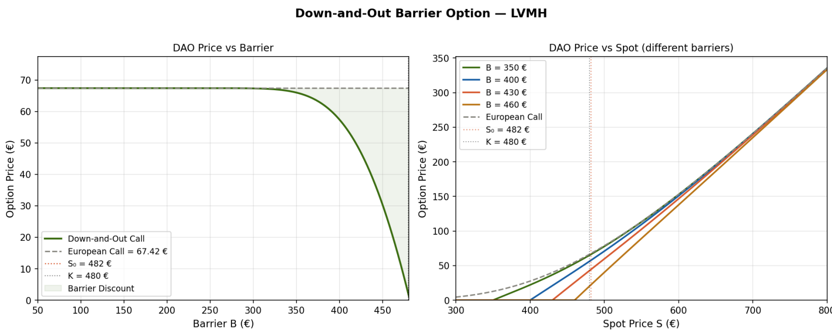

Figure 6: DAO Barrier Option - LVMH.

This figure illustrates the pricing dynamics of the *Down-and-Out* option. The left graph highlights the "barrier discount": the closer the required barrier is to the spot price, the higher the knock-out risk, heavily reducing the option's premium. Conversely, the right graph demonstrates that as the underlying asset's price increases, the distance to the barrier widens, thereby mitigating the knock-out risk. Consequently, the value of the DAO option naturally converges toward the theoretical price of a standard European Call, as the probability of hitting the barrier becomes statistically negligible.

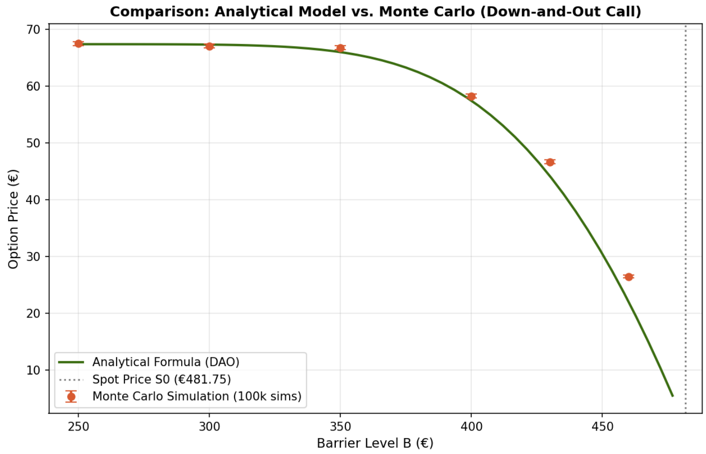

Figure 7: Monte Carlo vs Analytical - DAO.

We also put the graph of the comparison between the Monte Carlo and Analytical simulation confirming our results.

5 Model Limitations and Discussion

While the Black–Scholes framework provides solutions for European and barrier options, it relies on a set of idealized assumptions that limit its applicability under real-world market conditions.

5.1 Model Limitations

Constant Volatility and the Volatility Smile

The primary limitation of the Black–Scholes model lies in the assumption of constant volatility. In reality, volatility varies with the strike price and maturity (the volatility smile). Thus, depending on the specific stock and/or call option, volatility can fluctuate significantly, as opposed to the ”flat” volatility profile assumed by Black–Scholes.

Path Continuity and Absence of Jumps

The geometric Brownian motion model also imposes an assumption of continuity. In practice, however, due to daily market closures or unforeseen events, the underlying asset may be subject to sudden price jumps. This can cause the asset to breach a barrier instantaneously, a phenomenon that is not captured by our model.

Constant Interest Rates and Dividends

The model assumes that the risk-free rate r remains constant throughout the life of the option. This becomes unrealistic for long-term maturities, where rates can evolve significantly and thus influence the price of the underlying asset.

5.2 Discussion

To face these limitations, new approaches have been developed. The first one, the CEV model [3], uses a local volatility, where volatility depends on time and the current level of the asset. The second one is the Heston model [4], which uses stochastic volatility, where volatility is treated as a random process. These models give a more realistic picture of how markets behave.

Conclusion

In this project, we worked on the valuation of exotic options, especially barrier options, which are usually complex to handle in financial mathematics. We observed, for example, that adding a deactivation clause, like in an "up-and-out" call option, strongly influences how hedging strategies are applied in the Black-Scholes model.
Our study allowed us to highlight the importance of stochastic tools in financial mathematics. Indeed, the probabilistic approach, based on Girsanov's theorem and the reflection principle, allowed us to calculate the joint density of a Brownian motion and its maximum, so we could finally calculate the price of an exotic option. On the other hand, the partial differential equation approach gives a more local and dynamic perspective, which is very useful for risk management in more concrete situations.
Then, thanks to the Feynman-Kac theorem, we were able to link these two perspectives, showing that they converge to the exact same solution. This is not just a theoretical result: in practice, it is crucial to make sure that the option price becomes exactly zero at the barrier to eliminate arbitrage opportunities on the market.
Overall, this project helped us understand that even if the mathematical framework can seem quite complex at first, it plays a key role in correctly pricing derivatives and managing risks, especially when market conditions become more unstable.
In closing, we would like to thank A. Joulin for his invaluable assistance throughout this project. We are delighted to have worked with him because, in addition to his scientific expertise, he was always there to support and advise us during its development.
This project allowed us to deepen our knowledge of financial mathematics and gain practical experience in modeling it.

References

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[2] Nicolas PRIVAULT. *Stochastic Analysis in Discrete and Continuous Time*. Lecture Notes, Nanyang Technological University, Singapore, August 2012.

[3] John RANDAL. *The Constant Elasticity of Variance Option Pricing Model*. Master of Science thesis in Statistics and Operations Research, Victoria University of Wellington, April 1998.

[4] Sergei MIKHAILOV and Ulrich NÖGEL. *Heston's Stochastic Volatility Model Implementation, Calibration and Some Extensions*. Wilmott Magazine, Technical Article, 2003.

Appendix

This appendix mainly consists of the Python files used for the simulations.

Reflection Principle

```python
import numpy as np
import matplotlib.pyplot as plt

# Parameters
T, N, a = 1.0, 2000, 1.0
t = np.linspace(0, T, N)
dt = T / N

# Simulation of Brownian Motion
B = np.cumsum(np.random.normal(0, np.sqrt(dt), N))
B[0] = 0
hit = np.where(B >= a)[0]

# Application of the Reflection Principle if the barrier is hit
if len(hit) > 0:
    idx = hit[0]
    X = np.copy(B)
    X[idx:] = 2*a - B[idx:]

    # Plotting results
    plt.plot(t, B, label='$B_t$')
    plt.plot(t[idx:], X[idx:], label='$X_t$', linestyle='--')
    plt.axhline(y=a, color='k', label='a')
    plt.title('Reflection Principle')
    plt.legend()
    plt.show()
```

Listing 1: Python script for the Reflection Principle simulation

Drifted Brownian Motion

```python
import numpy as np
import matplotlib.pyplot as plt

# Visual style configuration
plt.style.use('seaborn-v0_8-whitegrid')

# Simulation parameters
T = 1.0           # Total time
N = 1000          # Number of steps
```

```python
dt = T / N         # Time increment
mu = 0.5           # Drift coefficient
sigma = 0.2        # Volatility coefficient
t = np.linspace(0, T, N)

# Drifted Brownian Motion generation
# Dynamics: dXt = mu*dt + sigma*dWt
np.random.seed(42) # For reproducibility
dW = np.random.normal(0, np.sqrt(dt), N)
dW[0] = 0
W = np.cumsum(dW)
X = mu * t + sigma * W

# Plotting
plt.figure(figsize=(12, 5))

# Plot the linear drift (mu * t)
plt.plot(t, mu * t, color='#e74c3c', linestyle='--', linewidth=1, label=r'Drift ($\mu t$)')

# Plot the stochastic process Xt
plt.plot(t, X, color='#2c3e50', linewidth=1.5, label=r'Brownian Motion $X_t$ ($\mu=0.5, \sigma=0.2$)')

# Horizontal line at zero
plt.axhline(0, color='black', linewidth=1)

# Axis configuration and titles
plt.title('Simulation of a Drifted Brownian Motion', fontsize=14, fontweight='bold', pad=15)
plt.xlabel('Time ($t$)', fontsize=11)
plt.ylabel('Value ($X_t$)', fontsize=11)
plt.legend(loc='upper left', frameon=True)

plt.tight_layout()
plt.show()
```

Listing 2: Simulation of a Drifted Brownian Motion

UAO Barrier options

```python
"""
Pricing of an Up-and-Out Barrier Option     LVMH (MC.PA)

Real data as of April 15, 2026 (Euronext Paris):
  S0 = 481.75 (closing price)
```

```
      = 0.32      (annualized historical volatility ~32%, sharp YTD decline)
  r   = 0.025     (ECB risk-free rate)
      = -0.24     (observed 12-month drift)
  T   = 1.0       (maturity: 1 year)
  K   =  480        (ATM strike, slightly below current price)

  We will use the exact analytical formula g(t,x) from page 27 of the
  report based on Proposition 8.
"""

import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

# LVMH Parameters (Real data 04/15/2026)

S0    = 481.75   # Current spot price (   )
K     = 480.00   # Strike (ATM)
sigma = 0.32     # Annualized volatility
r     = 0.025    # Risk-free rate (ECB)
mu    = -0.24    # Observed annual drift (not used in RN pricing)
T     = 1        # Maturity (years)

# Core Functions

def delta_plus(s, tau, r, sigma):
    """  ^+ _  (s) = [ln(s) + (r +      /2)  ] / (       )      eq. (31) of the report"""
    return (np.log(s) + (r + 0.5 * sigma**2) * tau) / (sigma * np.sqrt(tau))

def delta_minus(s, tau, r, sigma):
    """  ^- _  (s) = [ln(s) + (r -      /2)  ] / (       )      eq. (31) of the report"""
    return (np.log(s) + (r - 0.5 * sigma**2) * tau) / (sigma * np.sqrt(tau))

def bs_call(S, K, r, sigma, tau):
    """Standard European Call Price (Black-Scholes)"""
    if tau <= 0:
        return max(S - K, 0.0)
    d1 = delta_plus(S / K, tau, r, sigma)
    d2 = delta_minus(S / K, tau, r, sigma)
    return S * norm.cdf(d1) - K * np.exp(-r * tau) * norm.cdf(d2)

```

```
def uao_call(S, K, B, r, sigma, tau):
    """
    Up-and-Out Call Price     g(t, x) formula, page 27 of the report.

    g(t,x) = x [   (      (x/K)) -   (      (x/B)) ]
              - x (B/x)^(1+2r/    ) [   (      (B  /Kx)) -   (      (B/x)) ]
              - Ke^{-r  } [   (      (x/K)) -   (      (x/B)) ]
              + Ke^{-r  } (x/B)^(1-2r/    ) [   (      (B  /Kx)) -   (
   (B/x)) ]
    """
    if S >= B:
        return 0.0   # Knocked-out (deactivated)
    if tau <= 0:
        return max(S - K, 0.0) if S < B else 0.0

    alpha = 2.0 * r / sigma**2   # Exponent from change of measure

    dp = lambda s: delta_plus(s,  tau, r, sigma)
    dm = lambda s: delta_minus(s, tau, r, sigma)

    # Term 1: Spot component (S      (  ))
    A = S * (norm.cdf(dp(S/K)) - norm.cdf(dp(S/B)))

    # Term 2: Barrier correction on spot component
    C = S * (B/S)**(1 + alpha) * (norm.cdf(dp(B**2/(K*S))) - norm.cdf(
   dp(B/S)))

    # Term 3: Discounted strike component (Ke^{-r  }      (  ))
    E = np.exp(-r * tau) * K * (norm.cdf(dm(S/K)) - norm.cdf(dm(S/B)))

    # Term 4: Barrier correction on strike component
    G = np.exp(-r * tau) * K * (S/B)**(1 - alpha) * (
        norm.cdf(dm(B**2/(K*S))) - norm.cdf(dm(B/S))
    )

    return max(0.0, A - C - E + G)

# Numerical Application: LVMH

print("")
print("  Up-and-Out Barrier Option     LVMH")
print("  April 15, 2026")
print("")
print(f"   S   = {S0:.2f}        (Spot Price)")
print(f"  K   = {K:.2f}         (ATM Strike)")
print(f"      = {sigma*100:.0f}%         (Historical Volatility)")
print(f"  r   = {r*100:.1f}%        (ECB Risk-free Rate)")
```

```
print(f"      = {mu*100:.0f}%        (Observed Drift     info only)")
print(f"  T  = {T:.1f} year")
print()

barriers = [520, 540, 560, 580, 600, 650, 700, 800]
bs_price  = bs_call(S0, K, r, sigma, T)

print(f"  Standard European Call: {bs_price:.2f}    ")
print()
print(f"  {'Barrier B (   )':>16} | {'UAO Price (   )':>14} | {'Discount':>8} | {'Ratio UAO/BS':>12}")
print("  " + "-" * 58)
for B in barriers:
    p   = uao_call(S0, K, B, r, sigma, T)
    dec = (1 - p / bs_price) * 100 if bs_price > 0 else 0
    print(f"  {B:>16} | {p:>14.2f} | {dec:>7.1f}% | {p/bs_price:>12.3f}")

print()
print("  Note: the closer the barrier is to  S  ,")
print("  the higher the discount (high risk of deactivation).")

# Figure 1: UAO Price vs Barrier

fig, axes = plt.subplots(1, 2, figsize=(13, 5))
fig.suptitle("Up-and-Out Barrier Option     LVMH (MC.PA, 04/15/2026)",
             fontsize=13, fontweight='bold', y=1.01)

B_range = np.linspace(S0 + 5, 900, 400)
uao_prices = [uao_call(S0, K, B, r, sigma, T) for B in B_range]
bs_ref     = bs_call(S0, K, r, sigma, T)

ax = axes[0]
ax.plot(B_range, uao_prices, color='#185FA5', lw=2, label='Up-and-Out Call')
ax.axhline(bs_ref, color='#888780', lw=1.5, ls='--', label=f'European Call = {bs_ref:.2f}    ')
ax.axvline(S0, color='#D85A30', lw=1.2, ls=':', label=f' S   = {S0:.0f}    ')
ax.fill_between(B_range, uao_prices, bs_ref, alpha=0.08, color='#185FA5',
                label='Barrier Discount')
ax.set_xlabel("Barrier B (   )", fontsize=11)
ax.set_ylabel("Option Price (   )", fontsize=11)
ax.set_title("UAO Price vs Barrier", fontsize=11)
ax.legend(fontsize=9)
ax.grid(True, alpha=0.3)
```

```
ax.set_xlim(S0 + 5, 900)
ax.set_ylim(0, bs_ref * 1.15)

# Figure 2: UAO Price vs Spot (for different barriers)

ax2    = axes[1]
S_range = np.linspace(200, 480, 300)
colors  = ['#185FA5', '#3B6D11', '#D85A30', '#BA7517']
B_vals  = [520, 560, 620, 700]

for B, col in zip(B_vals, colors):
    prices = [uao_call(s, K, B, r, sigma, T) for s in S_range]
    ax2.plot(S_range, prices, color=col, lw=1.8, label=f'B = {B}    ')

bs_curve = [bs_call(s, K, r, sigma, T) for s in S_range]
ax2.plot(S_range, bs_curve, color='#888780', lw=1.5, ls='--', label='European Call')
ax2.axvline(S0, color='#D85A30', lw=1.2, ls=':', alpha=0.6, label=f'S   = {S0:.0f}    ')
ax2.axvline(K,  color='#000000', lw=1.0, ls=':', alpha=0.4, label=f'K = {K:.0f}    ')

ax2.set_xlabel("Spot Price S (   )", fontsize=11)
ax2.set_ylabel("Option Price (   )", fontsize=11)
ax2.set_title("UAO Price vs Spot (different barriers)", fontsize=11)
ax2.legend(fontsize=9)
ax2.grid(True, alpha=0.3)
ax2.set_xlim(200, 480)
ax2.set_ylim(0)

plt.tight_layout()
plt.savefig("lvmh_barrier_option.png",
            dpi=150, bbox_inches='tight')
print("\n  Figure saved: lvmh_barrier_option.png")
plt.close()

"""
Part- 2 : Simulation with the Monte Carlo Algorithm
"""

def uao_call_mc(S0, K, B, r, sigma, T, n_sims=100000, n_steps=252):
    """
    Monte Carlo pricing for an Up-and-Out Call.
    n_sims: Number of simulated price paths
    n_steps: Number of time steps per path (252 = daily observation)
    """
    dt = T / n_steps
```

```
    # Generate standard normal random variables for all steps and paths
    Z = np.random.standard_normal((n_steps, n_sims))

    # Pre-calculate drift and volatility for the log-process
    drift = (r - 0.5 * sigma**2) * dt
    vol = sigma * np.sqrt(dt)

    # Calculate daily log-returns
    increments = drift + vol * Z

    # Calculate cumulative sum to get log(S_t / S_0)
    log_paths = np.cumsum(increments, axis=0)

    # Transform log paths back to price paths (S_t)
    S_paths = S0 * np.exp(log_paths)

    # Find the maximum price reached during each simulation path
    # We include S0 in the maximum calculation
    S_max = np.maximum(S0, np.max(S_paths, axis=0))

    # Get the final prices at maturity T
    S_T = S_paths[-1, :]

    # Compute payoff: max(S_T - K, 0)
    payoffs = np.maximum(S_T - K, 0)

    # Apply Barrier condition: Payoff is 0 if the max price crossed the
     barrier B
    payoffs[S_max >= B] = 0.0

    # Discount payoffs to present value
    discounted_payoffs = np.exp(-r * T) * payoffs

    # Calculate Monte Carlo price and Standard Error
    price_mc = np.mean(discounted_payoffs)
    se = np.std(discounted_payoffs) / np.sqrt(n_sims)

    return price_mc, se

# --- 1. Data Preparation ---
# Varying the barrier B from  500  to  800
barriers_range = np.linspace(500, 800, 50)
analytical_prices = [uao_call(S0, K, b, r, sigma, T) for b in
    barriers_range]

# Choosing specific points for Monte Carlo (to avoid cluttering the
```

```
    graph)
mc_barriers = [520, 550, 600, 650, 700, 750, 800]
mc_results = []
mc_errors = []

print("Calculating Monte Carlo points...")
for b in mc_barriers:
    price, se = uao_call_mc(S0, K, b, r, sigma, T, n_sims=100000)
    mc_results.append(price)
    mc_errors.append(se)

# --- 2. Comparative Graph Creation ---
plt.figure(figsize=(10, 6))

# Analytical formula curve
plt.plot(barriers_range, analytical_prices, color='#185FA5', lw=2,
    label='Analytical Formula (Reference)')

# Monte Carlo points with error bars
plt.errorbar(mc_barriers, mc_results, yerr=mc_errors, fmt='o', color='#
    D85A30',
             ecolor='#D85A30', elinewidth=2, capsize=5, label='Monte
    Carlo Simulation (100k sims)')

# Aesthetics
plt.axvline(S0, color='black', linestyle=':', alpha=0.5, label=f'Spot
    Price S0 (  {S0})')
plt.title("Comparison: Analytical Model vs. Monte Carlo Simulation",
    fontsize=12, fontweight='bold')
plt.xlabel("Barrier Level B (  )", fontsize=11)
plt.ylabel("Option Price (  )", fontsize=11)
plt.legend()
plt.grid(True, alpha=0.3)

plt.savefig("method_comparison.png", dpi=150)
plt.show()
```

DAO Barrier options

```
"""
Pricing of a Down-and-Out Barrier Option    LVMH

Real data as of April 15, 2026 (Euronext Paris):
  S0  =  481 .75  (closing price)
      = 0.32      (annualized historical volatility ~32%, sharp YTD
    decline)
  r   = 0.025     (ECB risk-free rate)
```

```
      = -0.24     (observed 12-month drift)
  T   = 1.0        (maturity: 1 year)
  K   =  480        (ATM strike, slightly below current price)

  Exact analytical formula g(t,x)     DAO Proposal:

  g(t,x) = x [   (     (x/K)) -   (     (x/B)) ]
          - x (x/B)^(1 - 2r/    ) [   (     (B  /Kx)) -   (     (B/x)) ]
          - Ke^{-r  } [   (     (x/K)) -   (     (x/B)) ]
          + Ke^{-r  } (x/B)^(1 - 2r/    ) [   (     (B  /Kx)) -   ((B/x)) ]

  for 0 < B     x  (the option remains active as long as S stays above B).
"""

import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt

#           LVMH Parameters (Real data 04/15/2026)

S0    = 481.75   # Current spot price (   )
K     = 480.00   # Strike price (ATM)
sigma = 0.32     # Annualized volatility
r     = 0.025    # Risk-free rate (ECB)
mu    = -0.24    # Observed annual drift (not used in RN pricing)
T     = 1.0      # Maturity (years)

#           Core Functions

def delta_plus(s, tau, r, sigma):
    """    _    (s) = [ln(s) + (r +     /2)  ] / (       )"""
    return (np.log(s) + (r + 0.5 * sigma**2) * tau) / (sigma * np.sqrt(tau))

def delta_minus(s, tau, r, sigma):
    """    _    (s) = [ln(s) + (r -     /2)  ] / (       )"""
    return (np.log(s) + (r - 0.5 * sigma**2) * tau) / (sigma * np.sqrt(tau))

def bs_call(S, K, r, sigma, tau):
```

```
    """Standard European Call Price (Black-Scholes)"""
    if tau <= 0:
        return max(S - K, 0.0)
    d1 = delta_plus(S / K, tau, r, sigma)
    d2 = delta_minus(S / K, tau, r, sigma)
    return S * norm.cdf(d1) - K * np.exp(-r * tau) * norm.cdf(d2)

def dao_call(S, K, B, r, sigma, tau):
    """
    Formule analytique EXACTE pour un Call Down-and-Out (quand B <= K).
    Prix = Standard European Call - Down-and-In Call
    """
    if S <= B:
        return 0.0
    if tau <= 0:
        return max(S - K, 0.0)

    # Exposant alpha pour le principe de r flexion
    alpha = 2.0 * r / sigma**2

    dp = lambda s: (np.log(s) + (r + 0.5 * sigma**2) * tau) / (sigma * np.sqrt(tau))
    dm = lambda s: (np.log(s) + (r - 0.5 * sigma**2) * tau) / (sigma * np.sqrt(tau))

    # Termes A et E: Le Call Black-Scholes Classique
    A = S * norm.cdf(dp(S/K))
    E = np.exp(-r * tau) * K * norm.cdf(dm(S/K))

    # Termes C et G: La d cote de la barri re
    C = S * (B/S)**(1 + alpha) * norm.cdf(dp(B**2/(K*S)))

    # LA CORRECTION EST ICI : (B/S)**(alpha - 1) et non (1 - alpha) !
    G = np.exp(-r * tau) * K * (B/S)**(alpha - 1) * norm.cdf(dm(B**2/(K*S)))

    return max(0.0, A - E - C + G)

# Numerical Application: LVMH

print("")
print("  Down-and-Out Barrier Option     LVMH")
print("  April 15, 2026")
print("")
print(f"    S    = {S0:.2f}         (Spot Price)")
print(f"   K   = {K:.2f}         (ATM Strike)")
print(f"       = {sigma*100:.0f}%          (Historical Volatility)")
```

```python
print(f"  r   = {r*100:.1f}%        (ECB Risk-free Rate)")
print(f"      = {mu*100:.0f}%        (Observed Drift     info only)")
print(f"  T   = {T:.1f} year")
print()

# Tested barriers: all below S0 (DAO logic)
barriers = [460, 440, 420, 400, 380, 350, 300, 250]
bs_price  = bs_call(S0, K, r, sigma, T)

print(f"  Standard European Call: {bs_price:.2f}    ")
print()
print(f"  {'Barrier B (   )':>16} | {'DAO Price (   )':>14} | {'Discount':>8} | {'Ratio DAO/BS':>12}")
print("  " + "-" * 58)
for B in barriers:
    p   = dao_call(S0, K, B, r, sigma, T)
    dec = (1 - p / bs_price) * 100 if bs_price > 0 else 0
    print(f"  {B:>16} | {p:>14.2f} | {dec:>7.1f}% | {p/bs_price:>12.3f}")

print()
print("  Note: the closer the barrier is to  S  ,")
print("  the higher the discount (high risk of deactivation).")

# Figure 1: DAO Price as a function of the barrier

fig, axes = plt.subplots(1, 2, figsize=(13, 5))
fig.suptitle("Down-and-Out Barrier Option     LVMH",
             fontsize=13, fontweight='bold', y=1.01)

# Barriers from 50 up to just below S0
B_range    = np.linspace(50, S0 - 1, 400)
dao_prices = [dao_call(S0, K, B, r, sigma, T) for B in B_range]
bs_ref     = bs_call(S0, K, r, sigma, T)

ax = axes[0]
ax.plot(B_range, dao_prices, color='#3B6D11', lw=2, label='Down-and-Out Call')
ax.axhline(bs_ref, color='#888780', lw=1.5, ls='--', label=f'European Call = {bs_ref:.2f}    ')
ax.axvline(S0, color='#D85A30', lw=1.2, ls=':', label=f' S   = {S0:.0f}    ')
ax.axvline(K,  color='#000000', lw=1.0, ls=':', alpha=0.5, label=f'K = {K:.0f}    ')
ax.fill_between(B_range, dao_prices, bs_ref, alpha=0.08, color='#3B6D11',
                label='Barrier Discount')
```

```
ax.set_xlabel("Barrier B (   )", fontsize=11)
ax.set_ylabel("Option Price (   )", fontsize=11)
ax.set_title("DAO Price vs Barrier", fontsize=11)
ax.legend(fontsize=9)
ax.grid(True, alpha=0.3)
ax.set_xlim(50, S0 - 1)
ax.set_ylim(0, bs_ref * 1.15)

# Figure 2: DAO Price as a function of Spot (for different barriers)

ax2      = axes[1]
# Plotting spot price above each barrier
S_range = np.linspace(300, 800, 400)
colors  = ['#3B6D11', '#185FA5', '#D85A30', '#BA7517']
B_vals  = [350, 400, 430, 460]

for B, col in zip(B_vals, colors):
    prices = [dao_call(s, K, B, r, sigma, T) for s in S_range]
    ax2.plot(S_range, prices, color=col, lw=1.8, label=f'B = {B}    ')

bs_curve = [bs_call(s, K, r, sigma, T) for s in S_range]
ax2.plot(S_range, bs_curve, color='#888780', lw=1.5, ls='--', label='
    European Call')
ax2.axvline(S0, color='#D85A30', lw=1.2, ls=':', alpha=0.6, label=f'
     S   = {S0:.0f}    ')
ax2.axvline(K,  color='#000000', lw=1.0, ls=':', alpha=0.4, label=f'K =
     {K:.0f}    ')

ax2.set_xlabel("Spot Price S (   )", fontsize=11)
ax2.set_ylabel("Option Price (   )", fontsize=11)
ax2.set_title("DAO Price vs Spot (different barriers)", fontsize=11)
ax2.legend(fontsize=9)
ax2.grid(True, alpha=0.3)
ax2.set_xlim(300, 800)
ax2.set_ylim(0)

plt.tight_layout()
plt.savefig("lvmh_dao_barrier_option.png",
            dpi=150, bbox_inches='tight')
print("\n  Figure saved: lvmh_dao_barrier_option.png")
plt.close()

"""
Part 2 Monte Carlo Simulation
"""

def dao_call_mc(S0, K, B, r, sigma, T, n_sims=100000, n_steps=252):
```

```
    """
    Monte Carlo pricing for a Down-and-Out Call.
    n_steps: 252 (daily observation)
    """
    dt = T / n_steps

    # Generate Gaussian shocks
    Z = np.random.standard_normal((n_steps, n_sims))

    # Calculate increments of the geometric Brownian motion
    drift = (r - 0.5 * sigma**2) * dt
    vol = sigma * np.sqrt(dt)
    increments = drift + vol * Z

    # Reconstruct price paths
    log_paths = np.cumsum(increments, axis=0)
    S_paths = S0 * np.exp(log_paths)

    # Find the MINIMUM price reached (Down logic)
    S_min = np.minimum(S0, np.min(S_paths, axis=0))

    # Final price at maturity
    S_T = S_paths[-1, :]

    # Calculate Payoff
    payoffs = np.maximum(S_T - K, 0)

    # Down-and-Out Condition: Payoff is 0 if the price touched or crossed B
    payoffs[S_min <= B] = 0.0

    # Discounting to present value
    discounted_payoffs = np.exp(-r * T) * payoffs

    price_mc = np.mean(discounted_payoffs)
    se = np.std(discounted_payoffs) / np.sqrt(n_sims)

    return price_mc, se

# --- 4. Data Preparation for the Graph ---

# Range of barriers (from   200   to just below S0)
barriers_range = np.linspace(250, S0 - 5, 50)
analytical_prices = [dao_call(S0, K, b, r, sigma, T) for b in barriers_range]

# Specific points for Monte Carlo
```

```python
mc_barriers = [250, 300, 350, 400, 430, 460]
mc_results = []
mc_errors = []

print("Calculating Monte Carlo points (DAO)...")
for b in mc_barriers:
    price, se = dao_call_mc(S0, K, b, r, sigma, T, n_sims=100000)
    mc_results.append(price)
    mc_errors.append(se)

# --- 5. Comparative Graph Creation ---

plt.figure(figsize=(10, 6))

# Analytical Curve
plt.plot(barriers_range, analytical_prices, color='#3B6D11', lw=2, label='Analytical Formula (DAO)')

# Monte Carlo Points
plt.errorbar(mc_barriers, mc_results, yerr=mc_errors, fmt='o', color='#D85A30',
             ecolor='#D85A30', elinewidth=2, capsize=5, label='Monte Carlo Simulation (100k sims)')

# Aesthetics and labels
plt.axvline(S0, color='black', linestyle=':', alpha=0.5, label=f'Spot Price S0 (  {S0})')
plt.title("Comparison: Analytical Model vs. Monte Carlo (Down-and-Out Call)", fontsize=12, fontweight='bold')
plt.xlabel("Barrier Level B (  )", fontsize=11)
plt.ylabel("Option Price (  )", fontsize=11)
plt.legend()
plt.grid(True, alpha=0.3)

# Save and show
plt.savefig("dao_method_comparison.png", dpi=150)
plt.show()
```